Figure 4.27 summarizes the analysis and synthesis formulas for these types of signals.

As we have already indicated several times. there are two time-domain characteristics that determine the type of signal spectrum we obtain. These are whether the time variable is continuous or discrete, and whether the signal is periodic or aperiodic. Let us briefly summarize the results of the previous sections.

Continuous-time signals have aperiodic spectra. A close inspection of the Fourier series and Fourier transform analysis formulas for continuous-time signals does not reveal any kind of periodicity in the spectral domain. This lack of periodicity is a consequence of the fact that the complex exponential $\exp(j2\pi Ft)$ is a function of the continuous variable t. and hence it is not periodic in F. Thus the frequency range of continuous-time signals extends from $F = 0$ to $F = \infty$.

Discrete-time signals have periodic spectra. Indeed. both the Fourier series and the Fourier transform for discrete-time signals are periodic with period $\omega = 2\pi$. As a result of this periodicity. the frequency range of discrete-time signals is finite and extends from $\omega = -\pi$ to $w = \pi$ radians, where $\omega = \pi$ corresponds to the highest possible rate of oscillation.

Periodic signals have discrete spectra. As we have observed, periodic signals are described by means of Fourier series. The Fourier series coefficients provide the "lines" that constitute tho discrete spectrum. The line spacing A F or Δf is equal to the inverse of the period T_p or N. respectively, in the time domain. That is. $A F = 1/T_p$ for continuous-time periodic signals and $Af = 1/N$ for discrete-time signals.

Aperiodic finite energy signals have continuous spectra. This property is a direct consequence of the fact that both $X(F)$ and $X(\omega)$ are functions of $\exp(j2\pi Ft)$ and $\exp(j\omega n)$, respectively. which are continuous functions of the variables F and ω. The continuity in frequency is necessary to break the harmony and thus create aperiodic signals.

In summary. we can conclude that periodicity *with "period" α in one domain automatically implies discretization with "spacing" of l/a in the other domain, and vice versa.*

If we keep in mind that "period" in the frequency domain means the frequency range. "spacing" in the time domain is the sampling period T, line spacing in the frequency domain is AF. then $a = T_p$ implies that $l/a = 1/T_p = AF$, $a = N$ implies that $Af = 1/N$, and $a = F_s$ implies that $T = 1/F_s$.

These time-frequency dualities are apparent from observation of Fig. 4.27. We stress, however, that the illustrations used in this figure do not correspond to any actual transform pairs. Thus any comparison among them should be avoided.

A careful inspection of Fig. 4.27 also reveals some mathematical symmetries and dualities among the several frequency analysis relationships. In particular,

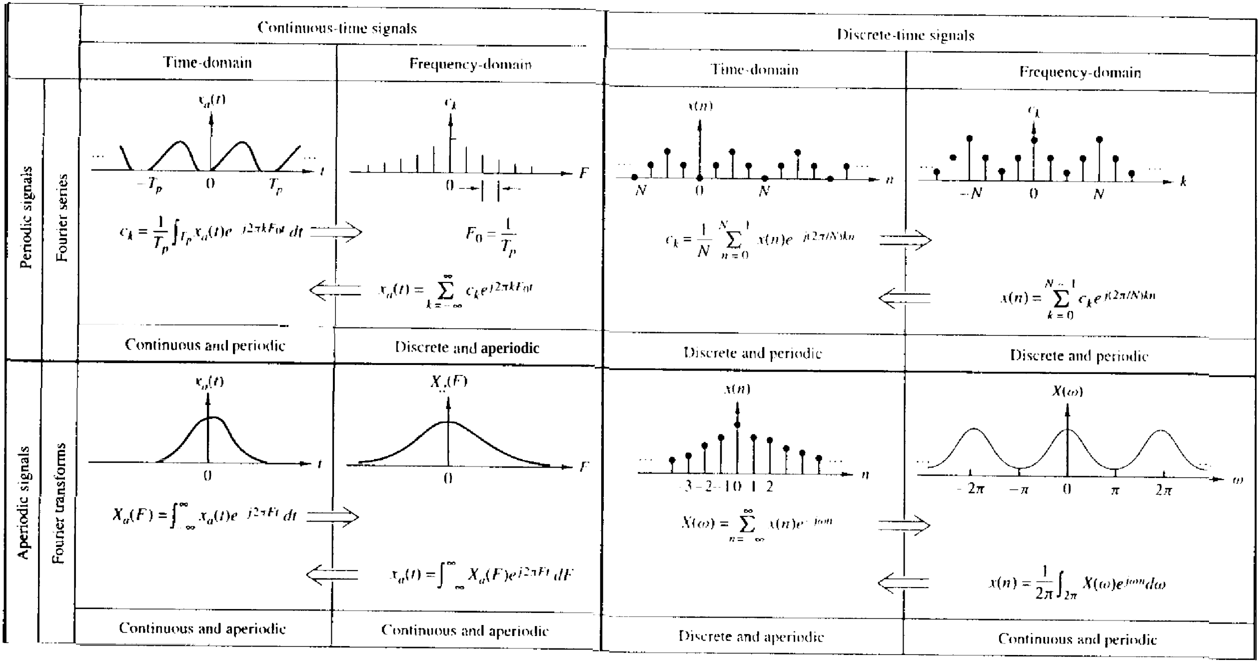


Figure 4.27 Summary of analysis and synthesis formulas.

we observe that there are dualities between the following analysis and synthesis equations:

1. The analysis and synthesis equations of the continuous-time Fourier transform.
2. The analysis and synthesis equations of the discrete-time Fourier series.
3. The analysis equation of the continuous-time Fourier series and the synthesis equation of the discrete-time Fourier transform.
4. The analysis equation of the discrete-time Fourier transform and the synthesis equation of the continuous-time Fourier series.

Note that all dual relations differ only in the sign of the exponent of the corresponding complex exponential. It is interesting to note that this change in sign can be thought of either as a folding of the signal or a folding of the spectrum, since

$$e^{-j2\pi Ft} = e^{j2\pi(-F)t} = e^{j2\pi F(-t)}$$

If we turn our attention now to the spectral density of signals, we recall that we have used the term ***energy density spectrum*** for characterizing finite-energy aperiodic signals and the term ***power density spectrum*** for periodic signals. This terminology is consistent with the fact that periodic signals are power signals and aperiodic signals with finite energy are energy signals.

4.3 PROPERTIES OF THE FOURIER TRANSFORM FOR DISCRETE-TIME SIGNALS

The Fourier transform for aperiodic finite-energy discrete-time signals described in the preceding section possesses a number of properties that are very useful in reducing the complexity of frequency analysis problems in many practical applications. In this section we develop the important properties of the Fourier transform. Similar properties hold for the Fourier transform of aperiodic finite-energy continuous-time signals.

For convenience, we adopt the notation

$$X(\omega) \equiv F\{x(n)\} = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \tag{4.3.1}$$

for the direct transform (analysis equation) and

$$x(n) \equiv F^{-1}\{X(\omega)\} = \frac{1}{2\pi}\int_{2\pi} X(\omega)e^{j\omega n}\,d\omega \tag{4.3.2}$$

for the inverse transform (synthesis equation). We also refer to $x(n)$ and $X(\omega)$ as a ***Fourier transform pair*** and denote this relationship with the notation

$$x(n) \xleftrightarrow{F} X(\omega) \tag{4.3.3}$$

Recall that $X(\omega)$ is periodic with period 2π. Consequently, any interval of length 2π is sufficient for the specification of the spectrum. Usually, we plot the spectrum in the fundamental interval $[-\pi, \pi]$. We emphasize that all the spectral information contained in the fundamental interval is necessary for the complete description or characterization of the signal. For this reason, the range of integration in (4.3.2) is always 2π, independent of the specific characteristics of the signal within the fundamental interval.

4.3.1 Symmetry Properties of the Fourier Transform

When a signal satisfies some symmetry properties in the time domain. these properties impose some symmetry conditions on its Fourier transform. Exploitation of any symmetry characteristics leads to simpler formulas for both the direct and inverse Fourier transform. A discussion of various symmetry properties and the implications of these properties in the frequency domain is given here.

Suppose that both the signal $x(n)$ and its transform $X(\omega)$ are complex-valued functions. Then they can be expressed in rectangular form as

$$x(n) = x_R(n) + jx_I(n) \tag{4.3.4}$$

$$X(\omega) = X_R(\omega) + jX_I(\omega) \tag{4.3.5}$$

By substituting (4.3.4) and $e^{-j\omega} = \cos\omega - j\sin\omega$ into **(4.3.1)** and separating the real and imaginary pans. we obtain

$$X_R(\omega) = \sum_{n=-\infty}^{\infty} [x_R(n)\cos\omega n + x_I(n)\sin\omega n] \tag{4.3.6}$$

$$X_I(\omega) = -\sum_{n=-\infty}^{\infty} [x_R(n)\sin\omega n - x_I(n)\cos\omega n] \tag{4.3.7}$$

In a similar manner, by substituting (4.3.5) and $e^{j\omega} = \cos\omega + j\sin\omega$ into (4.3.2), we obtain

$$x_R(n) = \frac{1}{2\pi}\int_{2\pi} [X_R(\omega)\cos\omega n - X_I(\omega)\sin\omega n]d\omega \tag{4.3.8}$$

$$x_I(n) = \frac{1}{2\pi}\int_{2\pi} [X_R(\omega)\sin\omega n + X_I(\omega)\cos\omega n]d\omega \tag{4.3.9}$$

Now, let us investigate some special cases.

Real signals. If $x(n)$ is real, then $x_R(n) = x(n)$ and $x_I(n) = 0$. Hence **(4.3.6)** and **(4.3.7)** reduce to

$$X_R(\omega) = \sum^{\infty} x(n)\cos\omega n \tag{4.3.10}$$

and

$$X_I(\omega) = -\sum_{n=-\infty}^{\infty} x(n)\sin \omega n \tag{4.3.11}$$

Since $\cos(-\omega n) = $ coswn and sin(−wn) = $-\sin \omega n$, it follows from (4.3.10) and (4.3.11) that

$$X_R(-\omega) = X_R(\omega) \qquad \text{(even)} \tag{4.3.12}$$

$$X_I(-\omega) = -X_I(\omega) \qquad \text{(odd)} \tag{4.3.13}$$

If we combine (4.3.12) and (4.3.13) into a single equation. we have

$$X^*(\omega) = X(-\omega) \tag{4.3.14}$$

In this case **we** say that the spectrum of a real signal has *Hermitian* ***symmetry.***

With the aid of Fig. 4.28, we observe that the magnitude and phase spectra for real signals are

$$|X(\omega)| = \sqrt{X_R^2(\omega) + X_I^2(\omega)} \tag{4.3.15}$$

$$\measuredangle X|\omega| = \tan^{-1}\frac{X_I(\omega)}{X_R(\omega)} \tag{4.3.16}$$

As a consequence of (4.3.12) and (4.3.13), the magnitude and phase spectra also possess the symmetry properties

$$|X(\omega)| = |X(-\omega)| \qquad \text{(even)} \tag{4.3.17}$$

$$\measuredangle X(-\omega) = -\measuredangle X(\omega) \qquad \text{(odd)} \tag{4.3.18}$$

In the case of the inverse transform of a real-valued signal [i.e., $x(n) = x_R(n)$], (4.3.8) implies that

$$x(n) = \frac{1}{2\pi}\int_{2\pi}[X_R(\omega)\cos \omega n - X_I(\omega)\sin \omega n]d\omega \tag{4.3.19}$$

Since both products $X_R(\omega)\cos \omega n$ and $X_I(\omega)\sin \omega n$ are even functions of w, we have

$$x(n) = \frac{1}{\pi}\int_0^{\pi}[X_R(\omega)\cos \omega n - X_I(\omega)\sin \omega n]d\omega \tag{4.3.20}$$

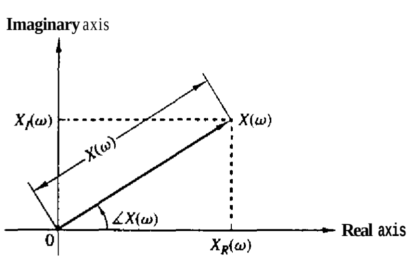


Figure 4.28 Magnitude and phase functions.

Real and even signals. If $x(n)$ is real and even [i.e., $x(-n) = x(n)$], then $x(n)\cos\omega n$ is even and $x(n)\sin\omega n$ is odd. Hence, from (4.3.10), (4.3.11), and (4.3.20) we obtain

$$X_R(\omega) = x(0) + 2\sum_{n=1}^{\infty} x(n)\cos\omega n \qquad \text{(even)} \tag{4.3.21}$$

$$X_I(\omega) = 0 \tag{4.3.22}$$

$$x(n) = \frac{1}{\pi}\int_0^{\pi} X_R(\omega)\cos\omega n\, d\omega \tag{4.3.23}$$

Thus real and even signals possess real-valued spectra. which, in addition, are even functions of the frequency variable w.

Real and odd signals. If $x(n)$ is real and odd [i.e., $x(-n) = -x(n)$], then $x(n)\cos\omega n$ is odd and $x(n)\sin\omega n$ is even. Consequently. (4.3.10). (4.3.11) and (4.3.20) imply that

$$X_R(\omega) = 0 \tag{4.3.24}$$

$$X_I(\omega) = -2\sum_{n=1}^{\infty} x(n)\sin\omega n \qquad \text{(odd)} \tag{4.3.25}$$

$$x(n) = -\frac{1}{\pi}\int_0^{\pi} X_I(\omega)\sin\omega n\, d\omega \tag{4.3.26}$$

Thus real-valued odd signals possess purely imaginary-valued spectral characteristics. which. in addition. are odd functions of the frequency variable ω.

Purely imaginary signals. In this case $x_R(n) = 0$ and $x(n) = jx_I(n)$. Thus (4.3.6), (4.3.7), and (4.3.9) reduce to

$$X_R(\omega) = \sum_{n=-\infty}^{\infty} x_I(n)\sin\omega n \qquad \text{(odd)} \tag{4.3.27}$$

$$X_I(\omega) = \sum_{n=-\infty}^{\infty} x_I(n)\cos\omega n \qquad \text{(even)} \tag{4.3.28}$$

$$x_I(n) = \frac{1}{\pi}\int_0^{\pi} [X_R(\omega)\sin\omega n + X_I(\omega)\cos\omega n]\, d\omega \tag{4.3.29}$$

If $x_I(n)$ is odd [i.e., $x_I(-n) = -x_I(n)$], then

$$X_R(\omega) = 2\sum_{n=1}^{\infty} x_I(n)\sin\omega n \qquad \text{(odd)} \tag{4.3.30}$$

$$X_I(\omega) = 0 \tag{4.3.31}$$

$$x_I(n) = \frac{1}{\pi}\int_0^{\pi} X_R(\omega)\sin\omega n\, d\omega \tag{4.3.32}$$

Similarly, if $x_I(n)$ is even [i.e., $x_I(-n) = x_I(n)$], we have

$$X_R(\omega) = 0 \tag{4.3.33}$$

$$X_I(\omega) = x_I(0) + 2\sum_{n=1}^{5} x_I(n)\cos\omega n \qquad \text{(even)} \tag{4.3.34}$$

$$x_I(n) = \frac{1}{\pi}\int_0^{\pi} X_I(\omega)\cos\omega n \, d\omega \tag{4.3.35}$$

An arbitrary, possibly complex-valued signal $x(n)$ can be decomposed as

$$\begin{aligned} x(n) = x_R(n) + jx_I(n) &= x_R^e(n) + x_R^o(n) + j[x_I^e(n) + x_I^o(n)] \\ &= x_e(n) + x_o(n) \end{aligned} \tag{4.3.36}$$

where, by definition,

$$x_e(n) = x_R^e(n) + jx_I^e(n) = \tfrac{1}{2}[x(n) + x^*(-n)]$$

$$x_o(n) = x_R^o(n) + jx_I^o(n) = \tfrac{1}{2}[x(n) - x^*(-n)]$$

The superscripts e and o denote the even and odd signal components, respectively. We note that $x_e(n) = x_e(-n)$ and $x_o(-n) = -x_o(n)$. From (4.3.36) and the Fourier transform properties established above, we obtain the following relationships:

$$\begin{aligned} x(n) &= [x_R^e(n) + jx_I^e(n)] + [x_R^o(n) + jx_I^o(n)] \\ &\quad\;\; \updownarrow \qquad\quad \updownarrow \qquad\qquad\quad \updownarrow \qquad\quad \updownarrow \\ X(\omega) &= [X_R^e(\omega) + jX_I^e(\omega)] + [X_R^o(\omega) + jX_I^o(\omega)] \end{aligned} \tag{4.3.37}$$

These symmetry properties of the Fourier transform are summarized in Table 4.4 and in Fig. **4.29.** They are often used to simplify Fourier transform calculations in practice.

Example 4.3.1

Determine and sketch $X_R(\omega)$, $X_I(\omega)$, $|X(\omega)|$, and $\measuredangle X(\omega)$ for the Fourier transform

$$X(\omega) = \frac{1}{1 - ae^{-j\omega}} \qquad -1 < a < 1 \tag{4.3.38}$$

Solution By multiplying both the numerator and denominator of (4.3.38) by the complex conjugate of the denominator, we obtain

$$X(\omega) = \frac{1 - ae^{j\omega}}{(1 - ae^{-j\omega})(1 - ae^{j\omega})} = \frac{1 - a\cos\omega - ja\sin\omega}{1 - 2a\cos\omega + a^2}$$

This expression can be subdivided into real and imaginary parts. Thus we obtain

$$X_R(\omega) = \frac{1 - a\cos\omega}{1 - 2a\cos\omega + a^2}$$

$$X_I(\omega) = -\frac{a\sin\omega}{1 - 2a\cos\omega + a^2}$$

Substitution of the last two equations into (4.3.15) and (4.3.16) yields the magnitude and phase spectra as

$$|X(\omega)| = \frac{1}{\sqrt{1 - 2a\cos\omega + a^2}} \tag{4.3.39}$$

TABLE 4.4 SYMMETRY PROPERTIES OF THE DISCRETE-TIME FOURIER TRANSFORM

Sequence	DTFT
$x(n)$	$X(\omega)$
$x^*(n)$	$X^*(-\omega)$
$x^*(-n)$	$X^*(\omega)$
$x_R(n)$	$X_e(\omega) = \frac{1}{2}[X(\omega) + X^*(-\omega)]$
$jx_I(n)$	$X_o(\omega) = \frac{1}{2}[X(\omega) - X^*(-\omega)]$
$x_e(n) = \frac{1}{2}[x(n) + x^*(-n)]$	$X_R(\omega)$
$x_o(n) = \frac{1}{2}[x(n) - x^*(-n)]$	$jX_I(\omega)$
Real Signals	
Any real signal $x(n)$	$X(\omega) = X^*(-\omega)$ $X_R(\omega) = X_R(-\omega)$ $X_I(\omega) = -X_I(-\omega)$ $\lvert X(\omega)\rvert = \lvert X(-\omega)\rvert$ $\measuredangle X(\omega) = -\measuredangle X(-\omega)$
$x_e(n) = \frac{1}{2}[x(n) + x(-n)]$ (real and even)	$X_R(\omega)$ (real and even)
$x_o(n) = \frac{1}{2}[x(n) - x(-n)]$ (real and odd)	$jX_I(\omega)$ (imaginary and odd)

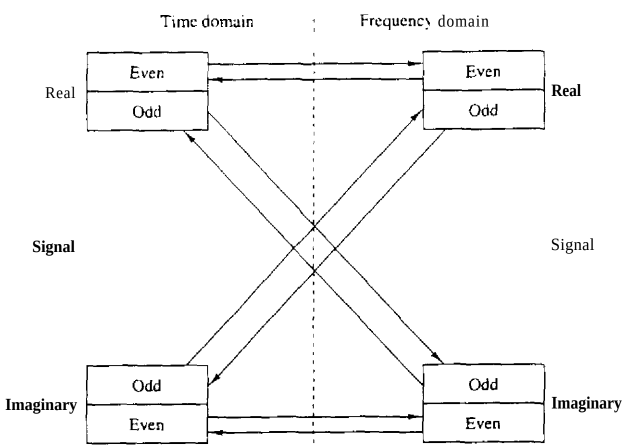


Figure 4.29 Summary of symmetry properties for the Fourier transform.

and

$$\measuredangle X(\omega) = -\tan^{-1} \frac{a \sin \omega}{1 - a \cos \omega} \tag{4.3.40}$$

Figures 4.30 and 4.31 show the graphical representation of these spectra for $a = 0.8$. The reader can easily verify that as expected. all symmetry properties for the spectra of real signals apply to this case.

Example 4.3.2

Determine the Fourier transform of the signal

$$x(n) = \begin{cases} A, & -M \le n \le M \\ 0, & \text{elsewhere} \end{cases} \tag{4.3.41}$$

Solution Clearly, $x(-n) = x(n)$. Thus $x(n)$ is a real and even signal. From (4.3.21) we obtain

$$X(\omega) = X_R(\omega) = A\left(1 + 2\sum_{n=1}^{M} \cos \omega n\right)$$

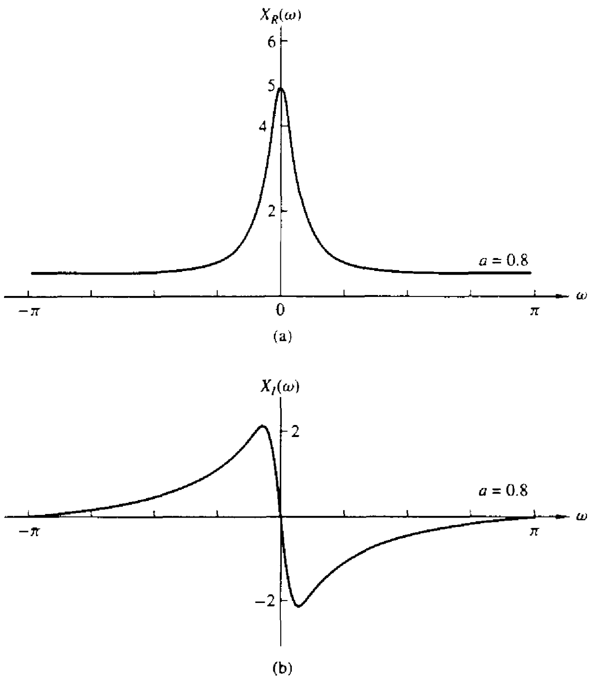


Figure 4.30 Graph of $X_R(\omega)$ and $X_I(\omega)$ for the transform in Example 4.3.1

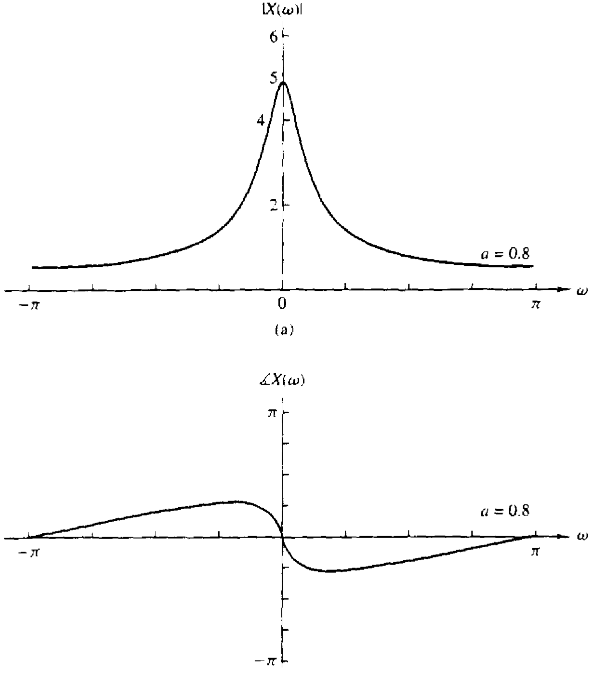


Figure 4.31 Magnitude and phase spectra of the transform in Example 4.3.1.

If we use the identity given in Problem 4.13, we obtain the simpler form

$$X(\omega) = A\frac{\sin(M + \frac{1}{2})\omega}{\sin(\omega/2)}$$

Since $X(\omega)$ is real, the magnitude and phase spectra are given by

$$|X(\omega)| = \left|A\frac{\sin(M + \frac{1}{2})\omega}{\sin(\omega/2)}\right| \tag{4.3.42}$$

and

$$\measuredangle X(\omega) = \begin{cases} 0, & \text{if } X(\omega) > 0 \\ \pi, & \text{if } X(\omega) < 0 \end{cases} \tag{4.3.43}$$

Figure 4.32 shows the graphs for $X(\omega)$.

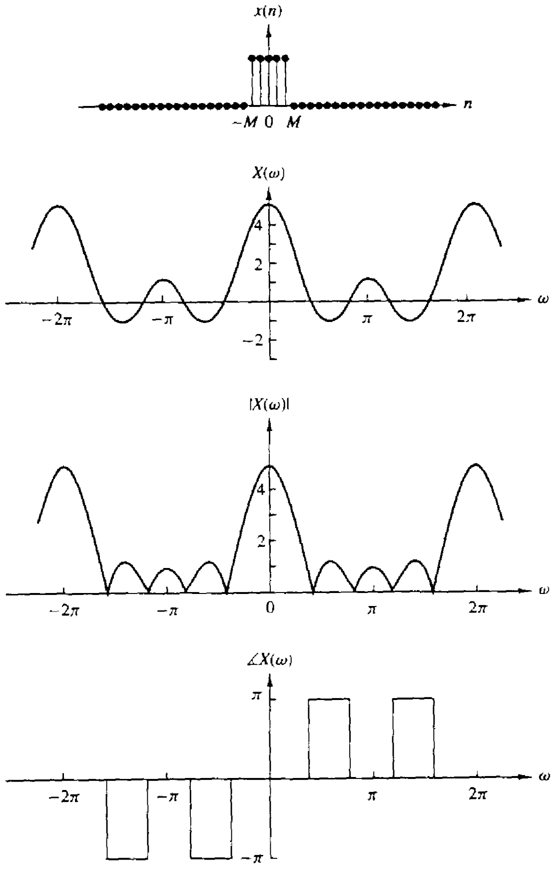


Figure 4.32 Spectral characteristics of rectangular pulse in Example 4.3.2.

4.3.2 Fourier Transform Theorems and Properties

In this section we introduce several Fourier transform theorems and illustrate their use in practice by examples.

Linearity. If

$$x_1(n) \overset{F}{\longleftrightarrow} X_1(\omega)$$

and

$$x_2(n) \overset{F}{\longleftrightarrow} X_2(\omega)$$

then

$$a_1x_1(n) + a_2x_2(n) \overset{F}{\longleftrightarrow} a_1X_1(\omega) + a_2X_2(\omega) \tag{4.3.44}$$

Simply stated, the Fourier transformation. viewed as an operation on a signal $x(n)$, is a linear transformation. Thus the Fourier transform of a linear combination of ***two*** or more signals is equal to the same linear combination of the Fourier transforms of the individual signals. This property is easily proved by using (4.3.1). The linearity property makes the Fourier transform suitable for the study of linear systems.

Example 4.3.3

Determine the Fourier transform of the signal

$$x(n) = a^{|n|} \qquad -1 < a < 1 \tag{4.3.45}$$

Solution First, we observe that $x(n)$ can be expressed as

$$x(n) = x_1(n) + x_2(n)$$

where

$$x_1(n) = \begin{cases} a^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

and

$$x_2(n) = \begin{cases} a^{-n}, & n < 0 \\ 0, & n \geq 0 \end{cases}$$

Beginning with the definition of the Fourier transform in (4.3.1). we have

$$X_1(\omega) = \sum_{n=-\infty}^{\infty} x_1(n)e^{-j\omega n} = \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \sum_{n=0}^{\infty}(ae^{-j\omega})^n$$

The summation is a geometric series that converges to

$$X_1(\omega) = \frac{1}{1 - ae^{-j\omega}}$$

provided that

$$|ae^{-j\omega}| = |a| \cdot |e^{-j\omega}| = |a| < 1$$

which is a condition that is satisfied in this problem. Similarly. the Fourier transform of $x_2(n)$ is

$$\begin{aligned} X_2(\omega) &= \sum_{n=-\infty}^{\infty} x_2(n)e^{-j\omega n} = \sum_{n=-\infty}^{-1} a^{-n}e^{-j\omega n} \\ &= \sum_{n=-\infty}^{-1} (ae^{j\omega})^{-n} = \sum_{k=1}^{\infty}(ae^{j\omega})^k \\ &= \frac{ae^{j\omega}}{1 - ae^{j\omega}} \end{aligned}$$

By combining these two transforms. we obtain the Fourier rransform of $x(n)$ in the form

$$
\begin{aligned}
X(\omega) &= X_1(\omega) + X_2(\omega) \\
&= \frac{1-a^2}{1-2a\cos\omega + a^2}
\end{aligned}
\tag{4.3.46}
$$

Figure 4.33 illustrates $x(n)$ and $X(\omega)$ for the case in which $a = 0.8$.

Time shifting. If

$$x(n) \overset{F}{\longleftrightarrow} X(\omega)$$

then

$$x(n-k) \overset{F}{\longleftrightarrow} e^{-j\omega k} X(\omega) \tag{4.3.47}$$

The proof of this property follows immediately from the Fourier transform of $x(n-k)$ by **making** a change in the summation index. Thus

$$
\begin{aligned}
F\{x(n-k)\} &= X(\omega)e^{-j\omega k} \\
&= |X(\omega)|e^{j[\measuredangle X(\omega) - \omega k]}
\end{aligned}
$$

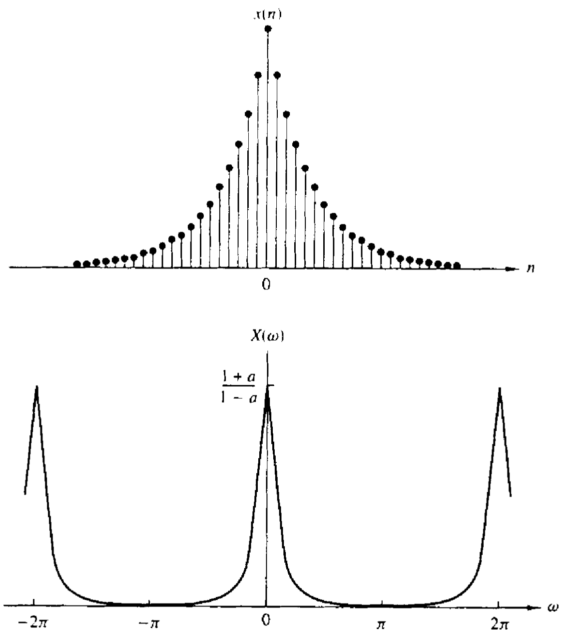


Figure 4.33 **Sequence** $x(n)$ **and its Fourier transform** in **Example 4.3.3** with $a = 0.8$.

This relation means that if a signal is shifted in the **time** domain by k samples, its magnitude spectrum remains unchanged. However, the phase spectrum is changed by an amount $-\omega k$. This result can easily be explained if we recall that the frequency content of a signal depends only on its shape. From a mathematical point of view, we can say that shifting by k in the time domain, is equivalent to multiplying the spectrum by $e^{-j\omega k}$ in the frequency domain.

Time reversal. If

$$x(n) \xleftrightarrow{F} X(\omega)$$

then

$$x(-n) \xleftrightarrow{F} X(-\omega) \tag{4.3.48}$$

This property can be established by performing the Fourier transformation of $x(-n)$ and making a simple change in the summation index. Thus

$$F\{x(-n)\} = \sum_{l=-\infty}^{\infty} x(l)e^{j\omega l} = X(-\omega)$$

If $x(n)$ is real, then from (4.3.17) and (4.3.18) we obtain

$$\begin{aligned} F\{x(-n)\} &= X(-\omega) = |X(-\omega)|e^{j\measuredangle X(-\omega)} \\ &= |X(\omega)|e^{-j\measuredangle X(\omega)} \end{aligned}$$

This means that if a signal is folded about the origin in time, its magnitude spectrum remains unchanged, and the phase spectrum undergoes a change in sign (phase reversal).

Convolution theorem. If

$$x_1(n) \xleftrightarrow{F} X_1(\omega)$$

and

$$x_2(n) \xleftrightarrow{F} X_2(\omega)$$

then

$$x(n) = x_1(n) * x_2(n) \xleftrightarrow{F} X(\omega) = X_1(\omega)X_2(\omega) \tag{4.3.49}$$

To prove (4.3.49), we recall the convolution formula

$$x(n) = x_1(n) * x_2(n) = \sum_{k=-\infty}^{\infty} x_1(k)x_2(n-k)$$

By multiplying both sides of this equation by the exponential $\exp(-j\omega n)$ and summing over all n, we obtain

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} = \sum_{n=-\infty}^{\infty}\left[\sum_{k=-\infty}^{\infty} x_1(k)x_2(n-k)\right]e^{-j\omega n}$$

After interchanging the order of the summations and making a simple change in the summation index, the right-hand side of this equation reduces to the product $X_1(\omega)X_2(\omega)$. Thus (4.3.49) is established.

The convolution theorem is one of the most powerful tools in linear systems analysis. That is. if we convolve two signals in the time domain. then this is equivalent to multiplying their spectra in the frequency domain. In later chapters we will see that the convolution theorem provides an important computational tool for many digital signal processing applications.

Example 4.3.4

By use of (4.3.49). determine the convolution of the sequences

$$x_1(n) = x_2(n) = \{1, \underset{\uparrow}{1}, 1\}$$

Solution By using (4.3.21). we obtain

$$X_1(\omega) = X_2(\omega) = 1 + 2\cos\omega$$

Then

$$\begin{aligned} X(\omega) &= X_1(\omega)X_2(\omega) = (1 + 2\cos\omega)^2 \\ &= 3 + 4\cos\omega + 2\cos 2\omega \\ &= 3 + 2(e^{j\omega} + e^{-j\omega}) + (e^{j2\omega} + e^{-j2\omega}) \end{aligned}$$

Hence the convolution of $x_1(n)$ with $x_2(n)$ is

$$x(n) = \{1\ 2\ \underset{\uparrow}{3}\ 2\ 1\}$$

Figure 4.34 illustrates the foregoing relationships.

The correlation theorem. If

$$x_1(n) \overset{F}{\longleftrightarrow} X_1(\omega)$$

and

$$x_2(n) \overset{F}{\longleftrightarrow} X_2(\omega)$$

then

$$r_{x_1x_2}(m) \overset{F}{\longleftrightarrow} S_{x_1x_2}(\omega) = X_1(\omega)X_2(-\omega) \tag{4.3.50}$$

The proof of (4.3.50) is similar to the proof of (4.3.49). In this case. we have

$$r_{x_1x_2}(n) = \sum_{k=-\infty}^{\infty} x_1(k)x_2(k-n)$$

By multiplying both sides of this equation by the exponential $\exp(-j\omega n)$ and summing over all n, we obtain

$$S_{x_1x_2}(\omega) = \sum_{n=-\infty}^{\infty} r_{x_1x_2}(n)e^{-j\omega n} = \sum_{n=-\infty}^{\infty}\left[\sum_{k=-\infty}^{\infty} x_1(k)x_2(k-n)\right]e^{-j\omega n}$$

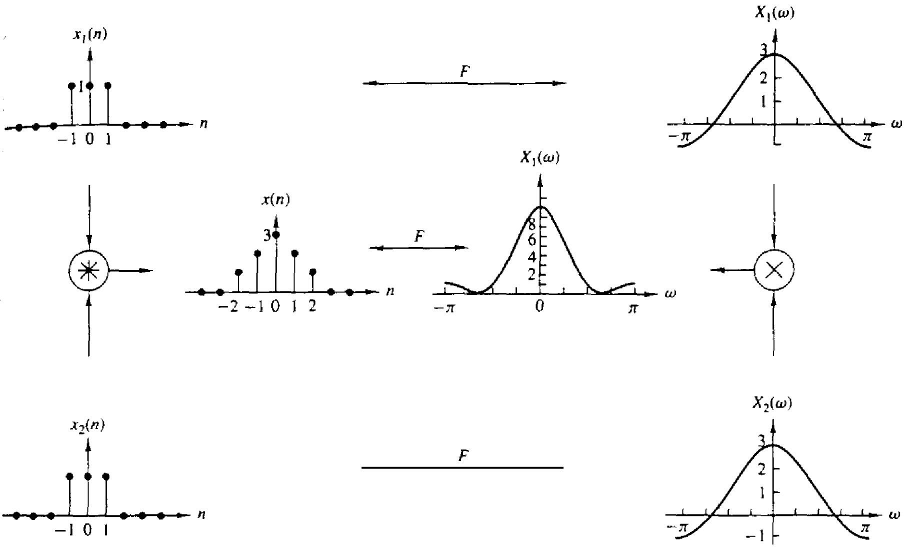


Figure 4.34 **Graphical representation of the convolution property.**

Finally, we interchange the order of the summations and make a change in the summation index. Thus we find that the right-hand side of the equation above reduces to $X_1(\omega)X_2(-\omega)$. The function $S_{x_1x_2}(\omega)$ is called the cross-energy *density* spectrum of the signals $x_1(n)$ and $x_2(n)$.

The Wiener–Khintchine theorem. Let $x(n)$ be a real signal. Then

$$r_{xx}(l) \overset{F}{\longleftrightarrow} S_{xx}(\omega) \tag{4.3.51}$$

That is, the energy spectral density of an energy signal is the Fourier transform of its autocorrelation sequence. This is a special case of (4.3.50).

This is a very important result. It means that the autocorrelation sequence of a signal and its energy spectral density contain the same information about the signal. Since, neither of these contains any phase information, it is impossible to uniquely reconstruct the signal from the autocorrelation function or the energy density spectrum.

Example 4 3 5

Determine the energy density spectrum of the signal

$$x(n) = a^n u(n) \qquad -1 < a < 1$$

Solution From Example 2.6.2 we found that the autocorrelation function for this signal is

$$r_{xx}(l) = \frac{1}{1-a^2}a^{|l|} \qquad \infty < l < \infty$$

By using the result in (4.3.46) for the Fourier transform of $a^{|l|}$, derived in Example 4.3.3, we have

$$F\{r_{xx}(l)\} = \frac{1}{1-a^2}F\{a^{|l|}\} = \frac{1}{1-2a\cos\omega + a^2}$$

Thus, according to the Wiener–Khintchine theorem,

$$S_{xx}(\omega) = \frac{1}{1-2a\cos\omega + a^2}$$

Frequency shifting. If

$$x(n) \stackrel{F}{\longleftrightarrow} X(\omega)$$

then

$$e^{j\omega_0 n}x(n) \stackrel{F}{\longleftrightarrow} X(\omega - \omega_0) \tag{4.3.52}$$

This property is easily proved by direct substitution into the analysis equation (4.3.1). According to this property, multiplication of a sequence $x(n)$ by $e^{j\omega_0 n}$ is equivalent to a frequency translation of the spectrum $X(\omega)$ by ω_0. This frequency translation is illustrated in Fig. 4.35. Since the spectrum $X(\omega)$ is periodic, the shift ω_0 applies to the spectrum of the signal in every period.

The modulation theorem. If

$$x(n) \stackrel{F}{\longleftrightarrow} X(\omega)$$

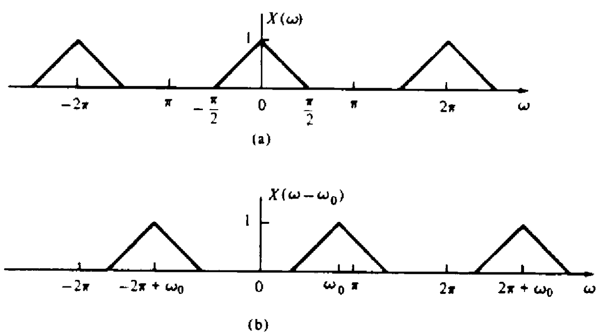


Figure 4.35 Illustration of the frequency-shifting property of the Fourier transform.

then

$$x(n)\cos\omega_0 n \overset{F}{\longleftrightarrow} \tfrac{1}{2}[X(\omega+\omega_0)+X(\omega-\omega_0)] \tag{4.3.53}$$

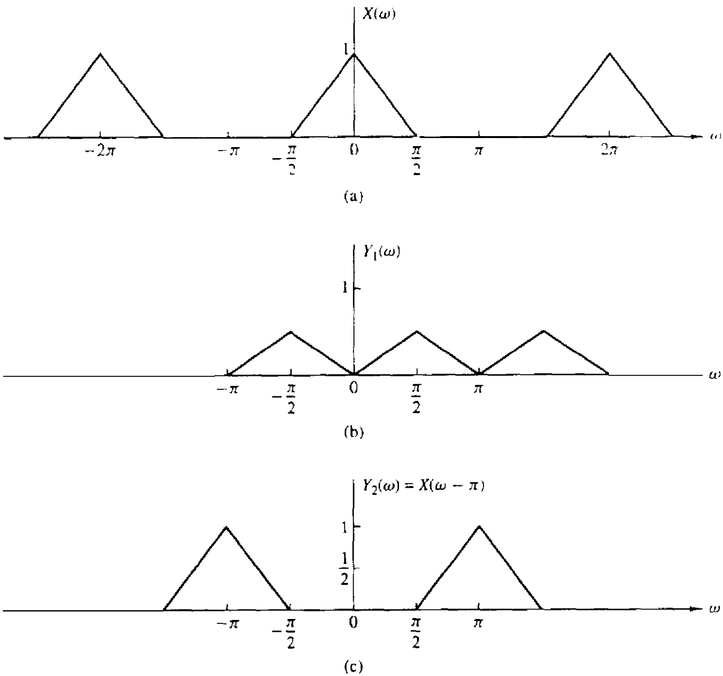
To prove the modulation theorem. we first express the signal $\cos\omega_0 n$ as

$$\cos\omega_0 n = \tfrac{1}{2}(e^{j\omega_0 n}+e^{-j\omega_0 n})$$

Upon multiplying $x(n)$ by these two exponentials and using the frequency-shiftinp property described in the preceding section, we obtain the desired result in (4.3.53).

Although the property given in (4.3.52) can also be viewed as (complex) modulation, in practice we prefer to use (4.3.53) because the signal $x(n)\cos\omega_0 n$ is real. Clearly. in this case the symmetry properties (4.3.12) and (4.3.13) are preserved.

The modulation theorem is illustrated in Fig. 4.36, which contains a plot of the spectra of the signals $x(n)$. $y_1(n) = x(n)\cos 0.5\pi n$ and $y_2(n) = x(n)\cos\pi n$.



Fire 4.36 **Graphical representation of the modulation** theorem.

Parseval's theorem. If

$$x_1(n) \xleftrightarrow{F} X_1(\omega)$$

and

$$x_2(n) \xleftrightarrow{F} X_2(\omega)$$

then

$$\sum_{n=-\infty}^{\infty} x_1(n)x_2^*(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X_1(\omega)X_2^*(\omega)d\omega \tag{4.3.54}$$

To prove this theorem, we use (4.3.1) to eliminate $X_1(\omega)$ on the right-hand side of (4.3.54). Thus we have

$$\frac{1}{2\pi}\int_{2\pi}\left[\sum_{n=-\infty}^{\infty} x_1(n)e^{-j\omega n}\right] X_2^*(\omega)d\omega$$

$$= \sum_{n=-\infty}^{\infty} x_1(n)\frac{1}{2\pi}\int_{2\pi} X_2^*(\omega)e^{-j\omega n}d\omega = \sum_{n=-\infty}^{\infty} x_1(n)x_2^*(n)$$

In the special case where $x_2(n) = x_1(n) = x(n)$, Parseval's relation (4.3.54) reduces to

$$\sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi}\int_{2\pi} |X(\omega)|^2 d\omega \tag{4.3.55}$$

We observe that the left-hand side of (4.3.55) is simply the energy E_x of the signal $x(n)$. It is also equal to the autocorrelation of $x(n)$, $r_{xx}(l)$, evaluated at $l = 0$. The integrand in the right-hand side of (4.3.55) is equal to the energy density spectrum, so the integral over the interval $-\pi \le \omega \le \pi$ yields the total signal energy. Therefore, we conclude that

$$E_x = r_{xx}(0) = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi}\int_{2\pi} |X(\omega)|^2 d\omega = \frac{1}{2\pi}\int_{-\pi}^{\pi} S_{xx}(\omega)d\omega \tag{4.3.56}$$

Multiplication of two sequences (Windowing theorem). If

$$x_1(n) \xleftrightarrow{F} X_1(\omega)$$

and

$$x_2(n) \xleftrightarrow{F} X_2(\omega)$$

then

$$x_3(n) \equiv x_1(n)x_2(n) \xleftrightarrow{F} X_3(\omega) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X_1(\lambda)X_2(\omega-\lambda)d\lambda \tag{4.3.57}$$

The integral on the right-hand side of (4.3.57) represents the convolution of the Fourier transforms $X_1(\omega)$ and $X_2(\omega)$. This relation is the dual of the time-domain convolution. In other words, the multipiication of two time-domain sequences is equivalent to the convolution of their Fourier transforms. On the other hand, the convolution of two time-domain sequences is equivalent to the multiplication of their Fourier transforms.

To prove (4.3.57) we begin with the Fourier transform of $x_3(n) = x_1(n)x_2(n)$ and use the formula for the inverse transform, namely,

$$x_1(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X_1(\lambda)e^{j\lambda n}\,d\lambda$$

Thus, we have

$$\begin{aligned}
X_3(\omega) &= \sum_{n=-\infty}^{\infty} x_3(n)e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x_1(n)x_2(n)e^{-j\omega n} \\
&= \sum_{n=-\infty}^{\infty}\left[\frac{1}{2\pi}\int_{-\pi}^{\pi} X_1(\lambda)e^{j\lambda n}\,d\lambda\right] x_2(n)e^{-j\omega n} \\
&= \frac{1}{2\pi}\int_{-\pi}^{\pi} X_1(\lambda)\,d\lambda\left[\sum_{n=-\infty}^{\infty} x_2(n)e^{-j(\omega-\lambda)n}\right] \\
&= \frac{1}{2\pi}\int_{-\pi}^{\pi} X_1(\lambda)X_2(\omega-\lambda)\,d\lambda
\end{aligned}$$

The convolution integral in (4.3.57) is known as the ***periodic convolution*** of $X_1(\omega)$ and $X_2(\omega)$ because it is the convolution of two periodic functions having the same period. We note that the limits of integration extend over a single period. Furthermore, we note that due to the periodicity of the Fourier transform for discrete-time signals, there is no "perfect" duality between the time and frequency domains with respect to the convolution operation, as in the case of continuous-time signals. Indeed, convolution in the time domain (aperiodic summation) is equivalent to multiplication of continuous periodic Fourier transforms. However, multiplication of aperiodic sequences is equivalent to periodic convolution of their Fourier transforms.

The Fourier transform pair in (4.3.57) will prove useful in our treatment of FIR filter design based on the window technique.

Differentiation in the frequency domain. If

$$x(n) \overset{F}{\longleftrightarrow} X(\omega)$$

then

$$nx(n) \overset{F}{\longleftrightarrow} j\frac{dX(\omega)}{d\omega} \tag{4.3.58}$$

To prove this property. we use the definition of the Fourier transform in (4.3.1) and differentiate the series term by term with respect to ω. Thus we obtain

$$\begin{aligned}\frac{dX(\omega)}{d\omega} &= \frac{d}{d\omega}\left[\sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}\right]\\ &= \sum_{n=-\infty}^{\infty} x(n)\frac{d}{d\omega}e^{-j\omega n}\\ &= -j\sum_{n=-\infty}^{\infty} nx(n)e^{-j\omega n}\end{aligned}$$

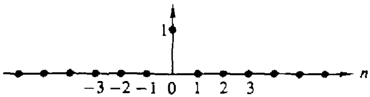
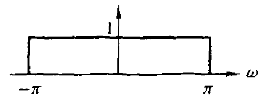
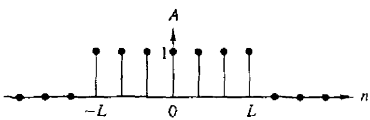
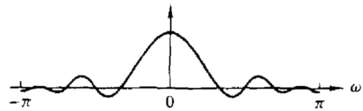
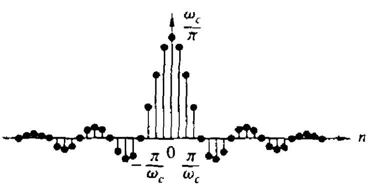
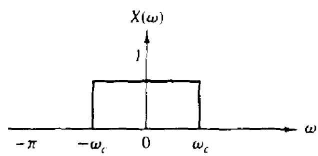
Now we multiply both sides of the equation by j to obtain the desired result in (4.3.58).

The properties derived in this section are summarized in Table 4.5. which serves as a convenient reference. Table 4.6 illustrates some useful Fourier transform pairs that will be encountered in later chapters.

TABLE 4.5 PROPERTIES OF THE FOURIER TRANSFORM FOR DISCRETE-TIME SIGNALS

Properly	Time Domain	Frequency Domain
Notation	$x(n)$	$X(\omega)$
	$x_1(n)$	$X_1(\omega)$
	$x_2(n)$	$X_2(\omega)$
Linearity	$a_1x_1(n) + a_2x_2(n)$	$a_1X_1(\omega) + a_2X_2(\omega)$
Time shifting	$x(n-k)$	$e^{-j\omega k}X(\omega)$
Time reversal	$x(-n)$	$X(-\omega)$
Convolution	$x_1(n) * x_2(n)$	$X_1(\omega)X_2(\omega)$
Correlation	$r_{x_1x_2}(l) = x_1(l) * x_2(-l)$	$S_{x_1x_2}(\omega) = X_1(\omega)X_2(-\omega)$ $= X_1(\omega)X_2^*(\omega)$ [if $x_2(n)$ is real]
Wiener–Khintchine theorem	$r_{xx}(l)$	$S_{xx}(\omega)$
Frequency shifting	$e^{j\omega_0 n}x(n)$	$X(\omega - \omega_0)$
Modulation	$x(n)\cos\omega_0 n$	$\frac{1}{2}X(\omega+\omega_0) + \frac{1}{2}X(\omega-\omega_0)$
Multiplication	$x_1(n)x_2(n)$	$\frac{1}{2\pi}\int_{-\pi}^{\pi} X_1(\lambda)X_2(\omega-\lambda)d\lambda$
Differentiation in the frequency domain	$nx(n)$	$j\frac{dX(\omega)}{d\omega}$
Conjugation	$x^*(n)$	$X^*(-\omega)$
Parseval's theorem	$\sum_{n=-\infty}^{\infty} x_1(n)x_2^*(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X_1(\omega)X_2^*(\omega)d\omega$	

TABLE 4.6 SOME USEFUL FOURIER TRANSFORM PAIRS FOR DISCRETE-TIME APERIODIC SIGNALS

Signal $x(n)$	Spectrum $X(\omega)$
 $x(n) = \delta(n)$	 $X(\omega) = 1$
 $x(n) = \begin{cases} A, & \lvert n \rvert \le L \\ 0, & \lvert n \rvert > L \end{cases}$	 $X(\omega) = A \dfrac{\sin\left(L + \frac{1}{2}\right)\omega}{\sin \frac{\omega}{2}}$
 $x(n) = \begin{cases} \dfrac{\omega_c}{\pi}, & n = 0 \\ \dfrac{\sin \omega_c n}{\pi n}, & n \ne 0 \end{cases}$	 $X(\omega) = \begin{cases} 1, & \lvert\omega\rvert < \omega_c \\ 0, & \omega_c \le \lvert\omega\rvert \le \pi \end{cases}$
$x(n) = \begin{cases} a^n, & n \ge 0 \\ 0, & n > 0 \end{cases}$	$X(\omega) = \dfrac{1}{1 - ae^{-j\omega}}$

4.4 FREQUENCY-DOMAIN CHARACTERISTICS OF LINEAR TIME-INVARIANT SYSTEMS

In this section we develop the characterization of linear time-invariant systems in the frequency domain. The basic excitation signals in this development are the complex exponentials and sinusoidal functions. The characteristics of the system are described by a function of the frequency variable ω called the frequency response, which is the Fourier transform of the impulse response $h(n)$ of the system.

The frequency response function completely characterizes a linear time-invariant system in the frequency domain. This allows us to determine the steady-state response of the system to any arbitrary weighted linear combination of sinusoids or complex exponentials. Since periodic sequences, in particular, lend themselves to a Fourier series decomposition as a weighted sum of harmonically related complex exponentials. it becomes a simple matter to determine the response of a linear time-invariant system to this class of signals. This methodology is also appiied to aperiodic signals since such signals can be viewed as a superposition of infinitesimal size complex exponentials.

4.4.1 Response to Complex Exponential and Sinusoidal Signals: The Frequency Response Function

In Chapter **2,** it was demonstrated that the response of any relaxed linear time-invariant system to an arbitrary input signal $x(n)$, is given by the convolution sum formula

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)x(n-k) \tag{4.4.1}$$

In this input-output relationship. the system is characterized in the time domain by its unit sample response $\{h(n), -\infty < n < \infty\}$.

To develop a frequency-domain characterization of the system, let us excite the system with the complex exponential

$$x(n) = Ae^{j\omega n} \qquad -\infty < n < \infty \tag{4.4.2}$$

where A is the amplitude and ω is any arbitrary frequency confined to the frequency interval $[-\pi, \pi]$. By substituting (4.4.2) into (4.4.1), we obtain the response

$$\begin{aligned} y(n) &= \sum_{k=-\infty}^{\infty} h(k)\left[Ae^{j\omega(n-k)}\right] \\ &= A\left[\sum_{k=-\infty}^{\infty} h(k)e^{-j\omega k}\right] e^{j\omega n} \end{aligned} \tag{4.4.3}$$

We observe that the term in brackets in **(4.4.3)** is a function of the frequency variable ω. In fact, this term is the Fourier transform of the unit sample response $h(k)$ of the system. Hence we denote this function as

$$H(\omega) = \sum_{k=-\infty}^{\infty} h(k)e^{-j\omega k} \tag{4.4.4}$$

Clearly, the function $H(\omega)$ exists if the system is BIBO stable, that is, if

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

With the definition in (4.4.4), the response of the system to the complex exponential given in **(4.4.2)** is

$$y(n) = AH(\omega)e^{j\omega n} \tag{4.4.5}$$

We note that the response is also in the form of a complex exponential with the same frequency as the input, but altered by the multiplicative factor $H(\omega)$.

As a result of this characteristic behavior. the exponential signal in (4.4.2) is called an eigenfunction of the system. In other words. an eigenfunction of a system is an input signal that produces an output that differs from the input by a constant multiplicative factor. The multiplicative factor is called an *eigenvalue* of the system. In this case, a complex exponential signal of the form (4.4.2) is an eigenfunction of a linear time-invariant system, and $H(\omega)$ evaluated at the frequency of the input signal is the corresponding eigenvalue.

Example 4.4.1

Determine the output sequence of the system with impulse response

$$h(n) = (\tfrac{1}{2})^n u(n) \tag{4.4.6}$$

when the input is the complex exponential sequence

$$x(n) = Ae^{j\pi n/2} \qquad -\infty < n < \infty$$

Solution First we evaluate the Fourier transform of the impulse response $h(n)$, and then we use (4.4.5) to determine $y(n)$. From Example 4.2.3 we recall that

$$H(\omega) = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n} = \frac{1}{1 - \frac{1}{2}e^{-j\omega}} \tag{4.4.7}$$

At $\omega = \pi/2$, (4.4.7) yields

$$H\left(\frac{\pi}{2}\right) = \frac{1}{1 + j\frac{1}{2}} = \frac{2}{\sqrt{5}}\, e^{-j26.6^\circ}$$

and therefore the output is

$$\begin{aligned} y(n) &= A\left(\frac{2}{\sqrt{5}}e^{-j26.6^\circ}\right)e^{j\pi n/2} \\ y(n) &= \frac{2}{\sqrt{5}}Ae^{j(\pi n/2 - 26.6^\circ)} \qquad -\infty < n < \infty \end{aligned} \tag{4.4.8}$$

This example clearly illustrates that the only effect of the system on the input signal is to scale the amplitude by $2/\sqrt{5}$ and shift the phase by -26.6°. Thus the output is also a complex exponential of frequency $\pi/2$, amplitude $2A/\sqrt{5}$, and phase -26.6°.

If we alter the frequency of the input signal. the effect of the system on the input also changes and hence the output changes. In particular. if the input sequence is a complex exponential of frequency π, that is,

$$x(n) = Ae^{j\pi n} \qquad -\infty < n < \infty \tag{4.4.9}$$

then, at $\omega = \pi$,

$$H(\pi) = \frac{1}{1 - \frac{1}{2}e^{-j\pi}} = \frac{1}{\frac{3}{2}} = \frac{2}{3}$$

and the output of the system is

$$y(n) = \tfrac{2}{3}Ae^{j\pi n} \qquad -\infty < n < \infty \tag{4.4.10}$$

We note that $H(\pi)$ is purely real [i.e., the phase associated with $H(\omega)$ is zero at $\omega = \pi$]. Hence. the input is scaled in amplitude by the factor $H(\pi) = \frac{2}{3}$, but the phase shift is zero.

In general. $H(\omega)$ is a complex-valued function of the frequency variable ω. Hence it can be expressed in polar form as

$$H(w) = |H(w)|e^{j\Theta(\omega)} \tag{4.4.11}$$

where $|H(\omega)|$ is the magnitude of $H(\omega)$ and

$$\Theta(\omega) = \measuredangle H(\omega)$$

which is the phase shift imparted on the input signal by the system at the frequency w.

Since $H(\omega)$ is the Fourier transform of $\{h(k)\}$, it follows that $H(\omega)$ is a periodic function with period 2π. Furthermore- we can view (4.4.4) as the exponential Fourier series expansion for $H(\omega)$, with $h(k)$ as the Fourier series coefficients. Consequently, the unit impulse $h(k)$ is related to $H(\omega)$ through the integral expression

$$h(k) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\omega)e^{j\omega k}\,d\omega \tag{4.4.12}$$

For a linear time-invariant system with a real-valued impulse response. the magnitude and phase functions possess symmetry properties which are developed as follows. From the definition of $H(\omega)$. we have

$$\begin{aligned} H(\omega) &= \sum_{k=-\infty}^{\infty} h(k)e^{-j\omega k} \\ &= \sum_{k=-\infty}^{\infty} h(k)\cos\omega k - j\sum_{k=-\infty}^{\infty} h(k)\sin\omega k \\ &= H_R(\omega) + jH_I(\omega) \\ &= \sqrt{H_R^2(\omega) + H_I^2(\omega)}\,e^{j\tan^{-1}[H_I(\omega)/H_R(\omega)]} \end{aligned} \tag{4.4.13}$$

where $H_R(\omega)$ and $H_I(\omega)$ denote the real and imaginary components of $H(\omega)$, defined *as*

$$\begin{aligned} H_R(\omega) &= \sum_{k=-\infty}^{\infty} h(k)\cos\omega k \\ H_I(\omega) &= -\sum_{k=-\infty}^{\infty} h(k)\sin\omega k \end{aligned} \tag{4.4.14}$$

It is clear from (4.4.12) that the magnitude and phase of $H(\omega)$, expressed in terms of $H_R(\omega)$ and $H_I(\omega)$, are

$$\begin{aligned} |H(\omega)| &= \sqrt{H_R^2(\omega) + H_I^2(\omega)} \\ \Theta(\omega) &= \tan^{-1}\frac{H_I(\omega)}{H_R(\omega)} \end{aligned} \tag{4.4.15}$$

We note that $H_R(\omega) = H_R(-\omega)$ and $H_I(\omega) = -H_I(-\omega)$, so that $H_R(\omega)$ is an even function of ω and $H_I(\omega)$ is an odd function of ω. As a consequence, it follows that $|H(\omega)|$ is an even function of ω and $\Theta(\omega)$ is an odd function of ω. Hence, if we know $|H(\omega)|$ and $\Theta(\omega)$ for $0 \le w \le \pi$, we also know these functions for $-\pi \le \omega \le 0$.

Example 4.4.2 Moving Average Filter

Determine the magnitude and phase of $H(\omega)$ for the three-point moving average (MA) system

$$y(n) = \tfrac{1}{3}[x(n+1) + x(n) + x(n-1)]$$

and plot these two functions for $0 \le \omega \le \pi$.

Solution Since

$$h(n) = \{\tfrac{1}{3}, \underset{\uparrow}{\tfrac{1}{3}}, \tfrac{1}{3}\}$$

it **follows** that

$$H(\omega) = \tfrac{1}{3}(e^{j\omega} + 1 + e^{-j\omega}) = \tfrac{1}{3}(1 + 2\cos\omega)$$

Hence

$$|H(\omega)| = \tfrac{1}{3}|1 + 2\cos\omega| \tag{4.4.16}$$

$$\Theta(\omega) = \begin{cases} 0, & 0 \le \omega \le 2\pi/3 \\ \pi, & 2\pi/3 \le \omega < \pi \end{cases}$$

Figure 4.37 illustrates the graphs of the magnitude and phase of $H(\omega)$. As indicated previously, $|H(\omega)|$ is an even function of frequency and $\Theta(\omega)$ is an odd function of

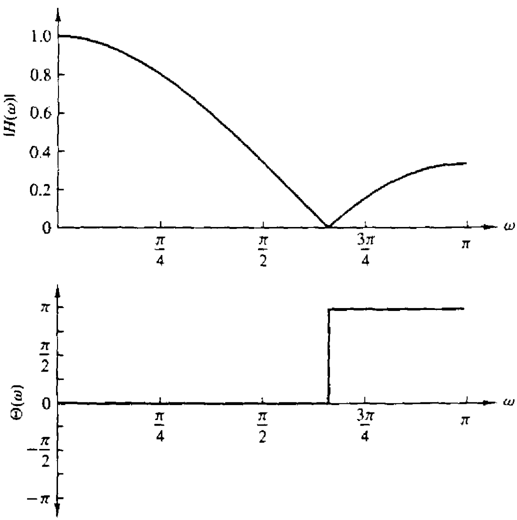


Figure 4.37 Magnitude and phase responses for the MA system in Example 4.4.2.

frequency. It is apparent from the frequency response characteristic $H(\omega)$ that this moving average filter smooths the input data. as we would expect from the input–output equarion.

The symmetry properties satisfied by the magnitude and phase functions of $H(\omega)$, and the fact that a sinusoid can be expressed as a sum or difference of two complex-conjugate exponential functions, imply that the response of a linear time-invariant system to a sinusoid is similar in form to the response when the input is a complex exponential. Indeed. if the input is

$$x_1(n) = Ae^{j\omega n}$$

the output is

$$y_1(n) = A|H(\omega)|e^{j\Theta(\omega)}e^{j\omega n}$$

On the other hand, if the input is

$$x_2(n) = Ae^{-j\omega n}$$

the response of the system is

$$\begin{aligned} y_2(n) &= A|H(-\omega)|e^{j\Theta(-\omega)}e^{-j\omega n} \\ &= A|H(\omega)|e^{-j\Theta(\omega)}e^{-j\omega n} \end{aligned}$$

where, in the last expression. we have made use of the symmetry properties $|H(\omega)| = |H(-\omega)|$ and $\Theta(\omega) = -\Theta(-\omega)$. Now. by applying the superposition property of the linear time-invariant system. we find that the response of the system to the input

$$x(n) = \tfrac{1}{2}[x_1(n) + x_2(n)] = A\cos\omega n$$

is

$$\begin{aligned} y(n) &= \tfrac{1}{2}[y_1(n) + y_2(n)] \\ y(n) &= A|H(\omega)|\cos[\omega n + \Theta(\omega)] \end{aligned} \tag{4.4.17}$$

Similarly. if the input is

$$x(n) = \frac{1}{j2}[x_1(n) - x_2(n)] = A\sin\omega n$$

the response of the system is

$$\begin{aligned} y(n) &= \frac{1}{j2}[y_1(n) - y_2(n)] \\ y(n) &= A|H(\omega)|\sin[\omega n + \Theta(\omega)] \end{aligned} \tag{4.4.18}$$

It is apparent from this discussion that $H(\omega)$, or equivalently. $|H(\omega)|$ and $\Theta(\omega)$, completely characterize the effect of the system on a sinusoidal input signal of any arbitrary frequency. Indeed. we note that $|H(\omega)|$ determines the amplification ($|H(\omega)| > 1$) or attenuation ($|H(\omega)| < 1$) imparted by the system on the input sinusoid. The phase $\Theta(\omega)$ determines the amount of phase shift imparted by the system on the input sinusoid. Consequently, by knowing $H(\omega)$, we are able to determine the response of the system to any sinusoidal input signal. Since $H(\omega)$ specifies the response of the system in the frequency domain, it is called the ***frequency response*** of the system. Correspondingly, $|H(\omega)|$ is called the ***magnitude response*** and $\Theta(\omega)$ is called the phase ***response*** of *the* system.

If the input to the system consists of more than one sinusoid, the superposition property of the linear system can be used to determine the response. The following examples illustrate the use of the superposition property.

Example 4.43

Determine the response of the system in Example **4.4.1** to the input signal

$$x(n) = 10 - 5\sin\frac{\pi}{2}n + 20\cos\pi n \qquad -\infty < n < \infty$$

Solution The frequency response of the system is given in (4.4.7) as

$$H(\omega) = \frac{1}{1 - \frac{1}{2}e^{-j\omega}}$$

The first term in the input signal is a fixed signal component corresponding to $\omega = 0$. Thus

$$H(0) = \frac{1}{1 - \frac{1}{2}} = 2$$

The second term in $x(n)$ has a frequency $\pi/2$. At this frequency the frequency response of the system is

$$H\left(\frac{\pi}{2}\right) = \frac{2}{\sqrt{5}}e^{-j26.6}$$

Finally, the third term in $x(n)$ has a frequency $w = \pi$. At this frequency

$$H(\pi) = \frac{2}{3}$$

Hence the response of the system to $x(n)$ is

$$y(n) = 20 - \frac{10}{\sqrt{5}}\sin\left(\frac{\pi}{2}n - 26.6^\circ\right) + \frac{40}{3}\cos\pi n \qquad -\infty < n < \infty$$

Example 4.44

A linear time-invariant system is described by the following difference equation:

$$y(n) = ay(n-1) + bx(n) \qquad 0 < a < 1$$

(a) Determine the magnitude and phase of the frequency response $H(\omega)$ of the system.

(b) Choose the parameter b so that the maximum value of $|H(\omega)|$ is unity, and sketch $|H(\omega)|$ and $\measuredangle H(\omega)$ for a = 0.9.

(c) Determine the output of the system to the input signal

$$x(n) = 5 + 12\sin\frac{\pi}{2}n - 20\cos\left(\pi n + \frac{\pi}{4}\right)$$

Solution The impulse response of the system is

$$h(n) = ba^n u(n)$$

Since $|a| < 1$. the system is BIBO stable and hence $H(\omega)$ exists.

(a) The frequency response is

$$\begin{aligned} H(\omega) &= \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n} \\ &= \frac{b}{1 - ae^{-j\omega}} \end{aligned}$$

Since

$$1 - ae^{-j\omega} = (1 - a\cos\omega) + ja\sin\omega$$

it follows that

$$\begin{aligned} |1 - ae^{-j\omega}| &= \sqrt{(1 - a\cos\omega)^2 + (a\sin\omega)^2} \\ &= \sqrt{1 + a^2 - 2a\cos\omega} \end{aligned}$$

and

$$\measuredangle(1 - ae^{-j\omega}) = \tan^{-1}\frac{a\sin\omega}{1 - a\cos w}$$

Therefore,

$$\begin{aligned} |H(\omega)| &= \frac{|b|}{\sqrt{1 + a^2 - 2a\cos\omega}} \\ \measuredangle H(\omega) &= \Theta(\omega) = \measuredangle b - \tan^{-1}\frac{a\sin\omega}{1 - a\cos\omega} \end{aligned}$$

(b) Since the parameter a is positive. the denominator of $|H(\omega)|$ attains a minimum at $\omega = 0$. Therefore. $|H(\omega)|$ attains its maximum value at $\omega = 0$. At this frequency we have

$$|H(0)| = \frac{|b|}{1 - a} = 1$$

which implies that $b = \pm(1 - a)$. We choose $b = 1 - a$. so that

$$|H(\omega)| = \frac{1 - a}{\sqrt{1 + a^2 - 2a\cos\omega}}$$

and

$$\Theta(\omega) = -\tan^{-1}\frac{a\sin\omega}{1 - a\cos\omega}$$

The frequency response plots for $|H(\omega)|$ and $\Theta(\omega)$ are illustrated in Fig. 4.38. We observe that this system attenuates high frequency signals.

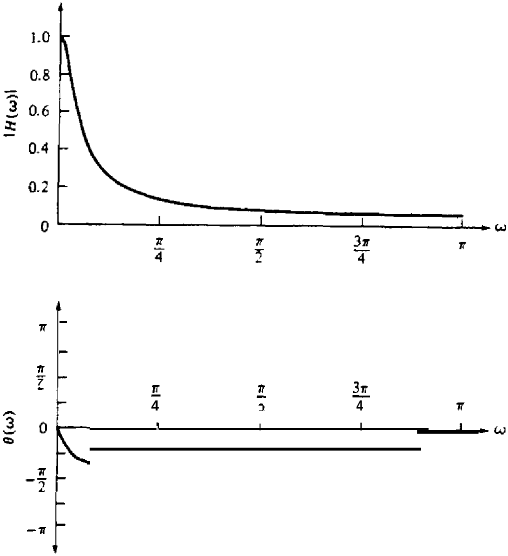


Figure 4.38 Magnitude and phase responses for the system in Example 4.4.4 with $a = 0.9$.

(c) The input signal consists of components of frequencies w = 0, $\pi/2$, and π. For $\omega = 0$, $|H(0)| = 1$ and $\Theta(0) = 0$. For $w = \pi/2$,

$$\left|H\left(\frac{\pi}{2}\right)\right| = \frac{1-a}{\sqrt{1+a^2}} = \frac{0.1}{\sqrt{1.81}} = 0.074$$

$$\Theta\left(\frac{\pi}{2}\right) = -\tan^{-1} a = -42^\circ$$

For $\omega = n$,

$$|H(\pi)| = \frac{1-a}{1+a} = \frac{0.1}{1.9} = 0.053$$

$$\Theta(\pi) = 0$$

Therefore, the **output** of the system is

$$\begin{aligned} y(n) &= 5|H(0)| + 12\left|H\left(\frac{\pi}{2}\right)\right| \sin\left[\frac{\pi}{2}n + \Theta\left(\frac{\pi}{2}\right)\right] \\ &\quad - 20|H(\pi)|\cos\left[\pi n + \frac{\pi}{4} + \Theta(\pi)\right] \\ &= 5 + 0.888\sin\left(\frac{\pi}{2}n - 42^\circ\right) - 1.06\cos\left(\pi n + \frac{\pi}{4}\right) \qquad -\infty < n < \infty \end{aligned}$$

In the most general case. if the input to the system consists of an arbitrary linear combination of sinusoids of the form

$$x(n) = \sum_{i=1}^{L} A_i \cos(\omega_i n + \phi_i) \qquad -\infty < n < \infty$$

where $\{A_i\}$ and $\{\phi_i\}$ are the amplitudes and phases of the corresponding sinusoidal components. then the response of the system is simply

$$y(n) = \sum_{i=1}^{L} A_i |H(\omega_i)| \cos[\omega_i n + \phi_i + \Theta(\omega_i)] \tag{4.4.19}$$

where $|H(\omega_i)|$ and $\Theta(\omega_i)$ are the magnitude and phase, respectively. imparted by the system to the individual frequency components of the input signal.

It is clear that depending on the frequency response $H(\omega)$ of the system, input sinusoids of different frequencies will be affected differently by the system. For example, some sinusoids may be completely suppressed by the system if $H(\omega) = 0$ at the frequencies of these sinusoids. Other sinusoids may receive no attenuation (or perhaps. some amplification) by the system. In effect. we can view the linear time-invariant system functioning as a filter to sinusoids of differenr frequencies, passing some of the frequency components to the output and suppressing or preventing other frequencv components from reaching the output. In fact, as discussed in Chapter 8. the basic digital filter design problem involves determining the parameters of a linear time-invariant system to achieve a desired frequency response $H(\omega)$.

4.4.2 Steady-State and Transient Response to Sinusoidal Input Signals

In the discussion in the preceding section. we determined the response of a linear time-invariant system to exponential and sinusoidal input signals applied to the system at $n = -\infty$. We usually call such signals eternal exponentials or eternal sinusoids, because they were applied at $n = -\infty$. In such a case, the response that we observe at the output of the system is the steady-state response. There is no transient response in this case,

On the other hand, if the exponential or sinusoidal signal is applied at some finite time instant. say at $n = 0$, the response of the system consists of two terms. the transient response and the steady-state response. To demonstrate this behavior, let us consider. as an example, the system described by the first-order difference equation

$$y(n) = ay(n-1) + x(n) \tag{4.4.20}$$

This system was considered in Section 2.4.2. Its response to any input $x(n)$ applied at $n = 0$ is given by *(2.4.8)* as

$$y(n) = a^{n+1} y(-1) + \sum_{k=0}^{n} a^k x(n-k) \qquad n \geq 0 \tag{4.4.21}$$

where $y(-1)$ is the initial condition.

Now, let us assume that the input to the system is the complex exponential

$$x(n) = Ae^{j\omega n} \qquad n \geq 0 \tag{4.4.22}$$

applied at $n = 0$. When we substitute **(4.4.22)** into (4.4.21), we obtain

$$\begin{aligned}
y(n) &= a^{n+1}y(-1) + A\sum_{k=0}^{n} a^k e^{j\omega(n-k)} \\
&= a^{n+1}y(-1) + A\left[\sum_{k=0}^{n}(ae^{-j\omega})^k\right]e^{j\omega n} \\
&= a^{n+1}y(-1) + A\frac{1 - a^{n+1}e^{-j\omega(n+1)}}{1 - ae^{-j\omega}}e^{j\omega n} \qquad n \geq 0 \\
&= a^{n+1}y(-1) - \frac{Aa^{n+1}e^{-j\omega(n+1)}}{1 - ae^{-j\omega}}e^{j\omega n} + \frac{A}{1 - ae^{-j\omega}}e^{j\omega n} \qquad n \geq 0
\end{aligned} \tag{4.4.23}$$

We recall that the system in **(4.4.20)** is BIBO stable if $|a| < 1$. In this case the two terms involving a^n+' in **(4.4.23)** decay toward zero as n approaches infinity. Consequentiy, we are left with the steady-state response

$$\begin{aligned}
y_{ss}(n) &= \lim_{n\to\infty} y(n) = \frac{A}{1 - ae^{-j\omega}}e^{j\omega n} \\
&= AH(\omega)e^{j\omega n}
\end{aligned} \tag{4.4.24}$$

The first two terms in **(4.4.23)** constitute the transient response of the system, that is,

$$y_{tr}(n) = a^n+'?(-1) - \frac{Aa^{n+1} \ -j\omega(n+1)}{1 - ae^{-j\omega}}e^{j\omega n} \qquad n \geq 0 \tag{4.4.25}$$

which decay toward zero as n approaches infinity. The first term in the transient response is the zero-input response of the system and the second term is the transient produced by the exponential input signal.

In general, all linear time-invariant BIBO systems behave in a similar fashion when excited by a complex exponential. or by a sinusoid at $n = 0$ or at some other finite time instant. That is, the transient response decays toward zero as $n \to \infty$, leaving only the steady-state response that we determined in the preceding section. In many practical applications, the transient response of the system is unimportant, and therefore it is usualty ignored in dealing with the response of the system to sinusoidal inputs.

4.4.3 Steady-State Response to Periodic Input Signals

Suppose that the input to a stable linear time-invariant system is a periodic signal $x(n)$ with fundamental period N. Since such a signal exists from $-\infty < n < \infty$, the total response of the system at any time instant n, is simply equal to the steady-state response.

To determine the response $y(n)$ of the svstem, we make use of the Fourier series representation of the periodic signal. which is

$$x(n) = \sum_{k=0}^{N-1} c_k e^{j2\pi kn/N} \qquad k = 0, 1, \ldots, N-1 \tag{4.4.26}$$

where the $\{c_k\}$ are the Fourier series coefficients. Now the response of the system to the complex exponential signal

$$x_k(n) = c_k e^{j2\pi kn/N} \qquad k = 0, 1, \ldots, N-1$$

is

$$y_k(n) = c_k H\left(\frac{2\pi}{N}k\right) e^{j2\pi kn/N} \qquad k = 0, 1, \ldots, N-1 \tag{4.4.27}$$

where

$$H\left(\frac{2\pi k}{N}\right) = H(\omega)|_{\omega=2\pi k/N} \qquad k = 0, 1, \ldots, N-1$$

By using the superposition principle for linear systems. we obtain the response of the system to the periodic signal $x(n)$ in (4.4.26) as

$$y(n) = \sum_{k=0}^{N-1} c_k H\left(\frac{2\pi k}{N}\right) e^{j2\pi kn/N} \qquad -\infty < n < \infty \tag{4.4.28}$$

This result implies that the response of the system to the periodic input signal $x(n)$ is also periodic with the same period N. The Fourier series coefficients for $y(n)$ are

$$d_k \equiv c_k H\left(\frac{2\pi k}{N}\right) \qquad k = 0, 1, \ldots, N-1 \tag{4.4.29}$$

Hence, the linear system can change the shape of the periodic input signal by scaling the amplitude and shifting the phase of the Fourier series components. but it does not affect the period of the periodic input signal.

4.4.4 Response to Aperiodic Input Signals

The convolution theorem. given in (4.3.49). provides the desired frequency-domain relationship for determining the output of an LTI system to an aperiodic finite-energy signal. If $\{x(n)\}$ denotes the input sequence. $\{y(n)\}$ denotes the output sequence. and $\{h(n)\}$ denotes the unit sample response of the system. then from the convolution theorem, we have

$$Y(\omega) = H(\omega)X(\omega) \tag{4.4.30}$$

where $Y(\omega)$, $X(\omega)$, and $H(\omega)$ are the corresponding Fourier transforms of $\{y(n)\}$. $\{x(n)\}$, and $\{h(n)\}$, respectively. From this relationship we observe that the spectrum of the output signal is equal to the spectrum of the input signal multiplied by the frequency response of the system.

If we express $Y(\omega)$, $H(\omega)$, and $X(\omega)$ in polar form, the magnitude and phase of the output signal can be expressed as

$$|Y(\omega)| = |H(\omega)||X(\omega)| \tag{4.4.31}$$

$$\measuredangle Y(\omega) = \measuredangle X(\omega) + \measuredangle H(\omega) \tag{4.4.32}$$

where $|H(\omega)|$ and $\measuredangle H(\omega)$ are the magnitude and phase responses of the system.

By its very nature, a finite-energy aperiodic signal contains a continuum of frequency components. The linear time-invariant system, through its frequency response function, attenuates some frequency components of the input signal and amplifies other frequency components. Thus the system acts as a ***filter*** to the input signal. Observation of the graph of $|H(\omega)|$ shows which frequency components are amplified and which are attenuated. On the other hand, the angle of $H(\omega)$ determines the phase shift imparted in the continuum of frequency components of the input signal as a function of frequency. If the input signal spectrum is changed by the system in an undesirable way, we say that the system has caused ***magnitude and phase distortion.***

We also observe that ***the output of a linear time-invariant system cannot contain frequency components that are not contained in the input signal.*** It takes either a linear time-variant system or a nonlinear system to create frequency components that are not necessarily contained in the input signal.

Figure 4.39 illustrates the time-domain and frequency-domain relationships that can be used in the analysis of BIBO-stable LTI systems. We observe that in time-domain analysis, we deal with the convolution of the input signal with the impulse response of the system to obtain the output sequence of the system. On the other hand, in frequency-domain analysis, we deal with the input signal spectrum $X(\omega)$ and the frequency response $H(\omega)$ of the system, which are related through multiplication, to yield the spectrum of the signal at the output of the system.

We can use the relation in (4.4.30) to determine the spectrum $Y(\omega)$ of the output signal. Then the output sequence $\{y(n)\}$ can be determined from the inverse Fourier transform

$$y(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} Y(\omega)e^{j\omega n}\,d\omega \tag{4.4.33}$$

However, this method is seldom used. Instead, the z-transform introduced in Chapter 3 is a simpler method for solving the problem of determining the output sequence $\{y(n)\}$.

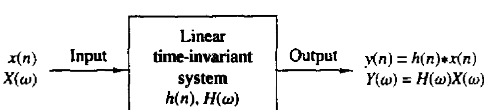


Figure 4.39 Time- and frequency-domain input-output relationships in LTI systems.

Let us return to the basic input-output relation in **(4.4.30)** and compute the squared magnitude of both sides. Thus we obtain

$$\begin{aligned} |Y(\omega)|^2 &= |H(\omega)|^2|X(\omega)|^2 \\ S_{yy}(\omega) &= |H(\omega)|^2 S_{xx}(\omega) \end{aligned} \tag{4.4.34}$$

where $S_{xx}(\omega)$ and $S_{yy}(\omega)$ are the energy density spectra of the input and output signals. respectively. By integrating **(4.4.34)** over the frequency range $(-\pi, n)$, we obtain the energy of the output signal as

$$\begin{aligned} E_y &= \frac{1}{2\pi}\int_{-\pi}^{\pi} S_{yy}(\omega)d\omega \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} |H(\omega)|^2 S_{xx}(\omega)d\omega \end{aligned} \tag{4.4.35}$$

Example 4.45

A linear time-invariant system is characterized by its impulse response

$$h(n) = (\tfrac{1}{2})^n u(n)$$

Determine the spectrum and the energy density spectrum of the output signal when the system is excited by the signal

$$x(n) = (\tfrac{1}{4})^n u(n)$$

Solution The frequency response function of the system

$$\begin{aligned} H(\omega) &= \sum_{n=0}^{\infty} (\tfrac{1}{2})^n e^{-j\omega n} \\ &= \frac{1}{1-\frac{1}{2}e^{-j\omega}} \end{aligned}$$

Similarly. the input sequence $\{x(n)\}$ has a Fourier transform

$$X(\omega) = \frac{1}{1-\frac{1}{4}e^{-j\omega}}$$

Hence the spectrum of the signal at the output of the system is

$$\begin{aligned} Y(\omega) &= H(\omega)X(\omega) \\ &= \frac{1}{(1-\frac{1}{2}e^{-j\omega})(1-\frac{1}{4}e^{-j\omega})} \end{aligned}$$

The corresponding energy density spectrum is

$$\begin{aligned} S_{yy}(\omega) &= |Y(\omega)|^2 = |H(\omega)|^2|X(\omega)|^2 \\ &= \frac{1}{(\frac{5}{4}-\cos\omega)(\frac{17}{16}-\frac{1}{2}\cos\omega)} \end{aligned}$$

4.4.5 Relationships Between the System Function and the Frequency Response Function

From the discussion in Section 4.2.6 we know that if the system function $H(z)$ converges on the unit circle. we can obtain the frequency response of the system by evaluating $H(z)$ on the unit circle. Thus

$$H(\omega) = H(z)|_{z=e^{j\omega}} = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n} \tag{4.4.36}$$

In the case where $H(z)$ is a rational function of the form $H(z) = B(z)/A(z)$. we have

$$H(\omega) = \frac{B(\omega)}{A(\omega)} = \frac{\sum_{k=0}^{M} b_k e^{-j\omega k}}{1 + \sum_{k=1}^{N} a_k e^{-j\omega k}} \tag{4.4.37}$$

$$= b_0 \frac{\prod_{k=1}^{M}(1 - z_k e^{-j\omega})}{\prod_{k=1}^{N}(1 - p_k e^{-j\omega})} \tag{4.4.38}$$

where the $\{a_k\}$ and $\{b_k\}$ are real. hut $\{z_k\}$ and $\{p_k\}$ may he complex-valued.

It is sometimes desirable to express the magnitude squared of $H(\omega)$ in terms of $H(z)$. First, we note that

$$|H(\omega)|^2 = H(\omega)H^*(\omega)$$

For the rational system function given by (4.4.38). we have

$$H^*(w) = b_0 \frac{\prod_{k=1}^{M}(1 - z_k^* e^{j\omega})}{\prod_{k=1}^{N}(1 - p_k^* e^{j\omega})} \tag{4.4.39}$$

It follows that $H^*(\omega)$ is obtained *by* evaluating $H^*(1/z^*)$ on the unit circle. where for a rational system function.

$$H^*(1/z^*) = b_0 \frac{\prod_{k=1}^{M}(1 - z_k^* z)}{\prod_{k=1}^{N}(1 - p_k^* z)} \tag{4.4.40}$$

However, when $\{h(n)\}$ is real or, equivalently. the coefficients $\{a_k\}$ and $\{b_k\}$ are real, complex-valued poles and zeros occur in complex-conjugate pairs. In this case. $H^*(1/z^*) = H(z^{-1})$. Consequently, $H^*(\omega) = H(-w)$.and

$$|H(\omega)|^2 = H(\omega)H^*(\omega) = H(\omega)H(-\omega) = H(z)H(z^{-1})|_{z=e^{j\omega}} \tag{4.4.41}$$

According to the correlation theorem for the z-transform (see Table 3.2), the function $H(z)H(z^{-1})$ is the z-transform of the autocorrelation sequence $\{r_{hh}(m)\}$ of the unit sample response $\{h(n)\}$. Then it follows from the Wiener–Khintchine theorem that $|H(\omega)|^2$ is the Fourier transform of $\{r_{hh}(m)\}$.

Similarly. if $H(z) = B(z)/A(z)$, the transforms $D(z) = B(z)B(z^{-1})$ and $C(z) = A(z)A(z^{-1})$ are the z-transforms of the autocorrelation sequences $\{c_l\}$ and $\{d_l\}$, where

$$c_l = \sum_{k=0}^{N-|l|} a_k a_{k+l} \qquad -N \le l \le N \tag{4.4.42}$$

$$d_l = \sum_{k=0}^{M-|l|} b_k b_{k+l} \qquad -M \le l \le M \tag{4.4.43}$$

Since the system parameters $\{a_k\}$ and $\{b_k\}$ are real valued, it follows that $c_l = c_{-l}$ and $d_l = d_{-l}$. By using this symmetry property, $|H(w)|^2$ may be expressed as

$$|H(\omega)|^2 = \frac{d_0 + 2\sum_{k=1}^{M} d_k \cos k\omega}{c_0 + 2\sum_{k=1}^{N} c_k \cos k\omega} \tag{4.4.44}$$

Finally, we note that coskw can be expressed as a polynomial function of $\cos w$. That is,

$$\cos kw = \sum_{m=0}^{k} \beta_m (\cos\omega)^m \tag{4.4.45}$$

where $\{\beta_m\}$ are the coefficients in the expansion. Consequentiy. the numerator and denominator of $|H(\omega)|^2$ can be viewed as polynomial functions of $\cos\omega$. The following example illustrates the foregoing relationships.

Example 4.4.6

Determine $|H(\omega)|^2$ for the **system**

$$y(n) = -0.1y(n-1) + 0.2y(n-2) + x(n) + x(n-1)$$

Solution The system function **is**

$$H(z) = \frac{1+z^{-1}}{1+0.1z^{-1} - 0.2z^{-2}}$$

and its ROC **is** $|z| > 0.5$. Hence $H(\omega)$ exists. Now

$$\begin{aligned} H(z)H(z^{-1}) &= \frac{1+z^{-1}}{1+0.1z^{-1}-0.2z^{-2}} \cdot \frac{1+z}{1+0.1z-0.2z^2} \\ &= \frac{2+z+z^{-1}}{1.05+0.08(z+z^{-1}) - 0.2(z^{-2}+z^{-2})} \end{aligned}$$

By evaluating $H(z)H(z^{-1})$ on the unit circle, we obtain

$$|H(\omega)|^2 = \frac{2+2\cos\omega}{1.05+0.16\cos\omega-0.4\cos 2\omega}$$

However, $\cos 2\omega = 2\cos^2\omega - 1$. Consequently, $|H(\omega)|^2$ may be expressed as

$$|H(\omega)|^2 = \frac{2(1+\cos\omega)}{1.45+0.16\cos\omega-0.8\cos^2\omega}$$

We note that given $H(z)$, it is straightforward to determine $H(z^{-1})$ and then $|H(\omega)|^2$. However, the inverse problem of determining $H(z)$ given $|H(\omega)|^2$ or the corresponding impulse response $\{h(n)\}$, is not straightforward. Since $|H(\omega)|^2$ does not contain the phase information in $H(\omega)$, it is not possible to uniquely determine $H(z)$.

To elaborate on the point, let us assume that the N poles and M zeros of $H(z)$ are $\{p_k\}$ and $\{z_k\}$, respectively. The corresponding poles and zeros of $H(z^{-1})$ are $\{1/p_k\}$ and $\{1/z_k\}$, respectively. Given $|H(\omega)|^2$ or, equivalently, $H(z)H(z^{-1})$, we can determine different system functions $H(z)$ by assigning to $H(z)$, a pole p_k or its reciprocal $1/p_k$, and a zero z_k or its reciprocal $1/z_k$. For example, if $N = 2$ and $M = 1$, the poles and zeros of $H(z)H(z^{-1})$ are $\{p_1, p_2, 1/p_1, 1/p_2\}$ and $\{z_1, 1/z_1\}$. If p_1 and p_2 are real, the possible poles for $H(z)$ are $\{p_1, p_2\}$, $\{1/p_1, 1/p_2\}$, $\{p_1, 1/p_2\}$, and $\{p_2, 1/p_1\}$ and the possible zeros are $\{z_1\}$ or $\{1/z_1\}$. Therefore, there are eight possible choices of system functions, all of which result in the same $|H(\omega)|^2$. Even if we restrict the poles of $H(z)$ to be inside the unit circle, there are still two different choices for $H(z)$, depending on whether we pick the zero $\{z_1\}$ or $\{1/z_1\}$. Therefore, we cannot determine $H(z)$ uniquely given only the magnitude response $|H(\omega)|$.

4.4.6 Computation of the Frequency Response Function

In evaluating the magnitude response and the phase response as functions of frequency, it is convenient to express $H(\omega)$ in terms of its poles and zeros. Hence we write $H(\omega)$ in factored form as

$$H(\omega) = b_0 \frac{\prod_{k=1}^{M}(1 - z_k e^{-j\omega k})}{\prod_{k=1}^{N}(1 - p_k e^{-j\omega k})} \tag{4.4.46}$$

or, equivalently, as

$$H(\omega) = b_0 e^{j\omega(N-M)} \frac{\prod_{k=1}^{M}(e^{j\omega} - z_k)}{\prod_{k=1}^{N}(e^{j\omega} - p_k)} \tag{4.4.47}$$

Let us express the complex-valued factors in **(4.4.47)** in polar form as

$$e^{j\omega} - z_k = V_k(\omega)e^{j\Theta_k(\omega)} \tag{4.4.48}$$

and

$$e^{j\omega} - p_k = U_k(\omega)e^{j\Phi_k(\omega)} \tag{4.4.49}$$

where

$$V_k(\omega) \equiv |e^{j\omega} - z_k|, \quad \Theta_k(\omega) \equiv \measuredangle(e^{j\omega} - z_k) \tag{4.4.50}$$

and

$$U_k(\omega) \equiv |e^{j\omega} - p_k|, \quad \Phi_k(\omega) = \measuredangle(e^{j\omega} - p_k) \tag{4.4.51}$$

The magnitude of $H(\omega)$ is equal to the product of magnitudes of all terms in **(4.4.47).** Thus, using (4.4.48) through (4.4.51), we obtain

$$|H(\omega)| = |b_0|\frac{V_1(\omega)\cdots V_M(\omega)}{U_1(\omega)U_2(\omega)\cdots U_N(\omega)} \tag{4.4.52}$$

since the magnitude of $e^{j\omega(N-M)}$ is 1.

The phase of $H(\omega)$ is the sum of the phases of the numerator factors, minus the phases of the denominator factors. Thus, by combining (4.4.48) through (4.4.51), we have

$$\begin{aligned}\measuredangle H(\omega) = \measuredangle b_0 + \omega(N-M) + \Theta_1(\omega) + \Theta_2(\omega) + \cdots + \Theta_M(\omega) \\ - [\Phi_1(\omega) + \Phi_2(\omega) + \cdots + \Phi_N(\omega)]\end{aligned} \tag{4.4.53}$$

The phase of the gain term b_0 is zero or π, depending on whether b_0 is positive or negative. Clearly. if we know the zeros and the poies of the system function $H(z)$, we can evaluate the frequency response from **(4.4.52)** and **(4.4.53).**

There is a geometric interpretation of the quantities appearing in **(4.4.52) and (4.4.53).** Let us consider a pole p_k and a zero z_k located at points A and B of the z-plane, as shown in Fig. 4.40(a). Assume that we wish to compute $H(\omega)$ at a specific value of frequency ω. The given value of ω determines the angle of $e^{j\omega}$ with the positive real axis. The tip of the vector $e^{j\omega}$ specifies a point L on the unit circle. The evaluation of the Fourier transform for the given value of ω is equivalent to evaluating the z-transform at the point L of the complex plane. Let us draw the vectors **AL** and **BL** from the pole and zero locations to the point L, at which we wish to compute the Fourier transform. From Fig. 4.40(a) it follows that

$$\mathbf{CL} = \mathbf{CA} + \mathbf{AL}$$

and

$$\mathbf{CL} = \mathbf{CB} + \mathbf{BL}$$

However, $\mathbf{CL} = e^{j\omega}$, $\mathbf{CA} = p_k$ and $\mathbf{CB} = z_k$. Thus

$$\mathbf{AL} = e^{j\omega} - p_k \tag{4.4.54}$$

and

$$\mathbf{BL} = e^{j\omega} - z_k \tag{4.4.55}$$

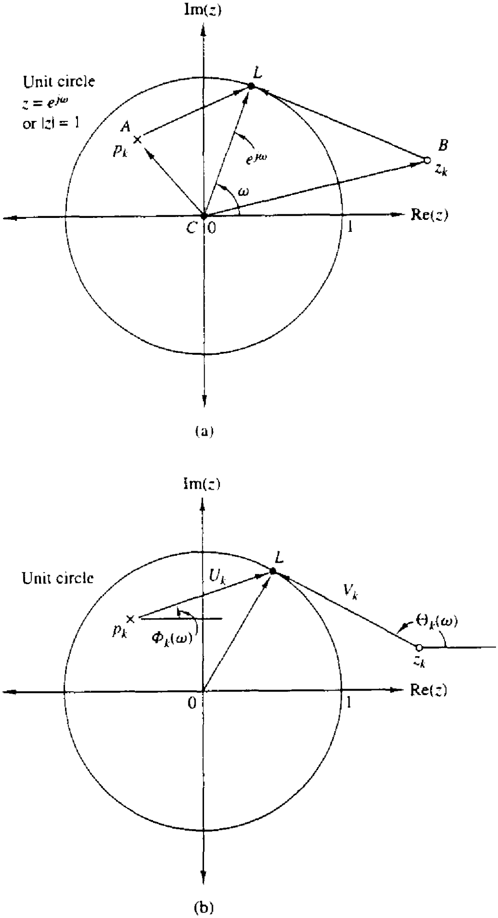


Figure 4.40 Geometric interpretation of the contribution of a pole and a zero to the Founer transform (1) magnitude: the factor V_k/U_k, (2) phase: the factor $\Theta_k - \Phi_k$.

By combining these relations with **(4.4.48)** and (4.4.49), we obtain

$$\mathbf{AL} = e^{j\omega} - p_k = U_k(\omega)e^{j\Phi_k(\omega)} \tag{4.4.56}$$

$$\mathbf{BL} = e^{j\omega} - z_k = V_k(\omega)e^{j\Theta_k(\omega)} \tag{4.4.57}$$

Thus $U_k(\omega)$ is the length of **AL**, that is, the distance of the pole p_k from the point L corresponding to $e^{j\omega}$, whereas $V_k(\omega)$ is the distance of the zero z_k from the same point L. The phases $\Phi_k(\omega)$ and $\Theta_k(\omega)$ are the angles of the vectors **AL** and **BL**

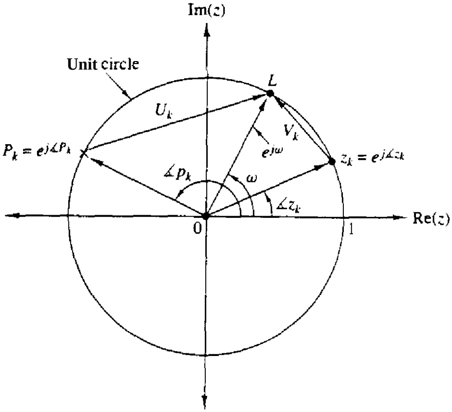


Figure 4.41 A zero on the unit circle causes $|H(\omega)| = 0$ and $w = \measuredangle z_k$. In contrast. a pole on the unit circle results in $|H(\omega)| = \infty$ at $\omega = \measuredangle p_k$.

with the positive real axis, respectively. These geometric interpretations are shown in Fig. 4.40(b).

Geometric interpretations are very useful in understanding how the location of poles and zeros affects the magnitude and phase of the Fourier transform. Suppose that a zero, say z_k, and a pole, say p_k, are on the unit circle as shown in Fig. 4.41. We note that at $\omega = \measuredangle z_k$, $V_k(\omega)$ and consequently $|H(\omega)|$ become zero. Similarly, at $\omega = \measuredangle p_k$ the length $U_k(\omega)$ becomes zero and hence $|H(\omega)|$ becomes infinite. Clearly, the evaluation of phase in these cases has no meaning.

From this discussion we can easily see that the presence of a zero close to the unit circle causes the magnitude of the frequency response, at frequencies that correspond to points of the unit circle close to that point, to be small. In contrast, the presence of a pole close to the unit circle causes the magnitude of the frequency response to be large at frequencies close to that point. Thus poles have the opposite effect of zeros. Also, placing a zero close to a pole cancels the effect of the pole, and vice versa. This can be also seen from (4.4.47), since if $z_k = p_k$, the terms $e^{j\omega} - z_k$ and $e^{j\omega} - p_k$ cancel. Obviously, the presence of both poles and zeros in a transform results in a greater variety of shapes for $|H(\omega)|$ and $\measuredangle H(\omega)$. This observation is very important in the design of digital filters. We conclude our discussion with the following example illustrating these concepts.

Example 4.4.7

Evaluate the frequency response of the system described by the system function

$$H(z) = \frac{1}{1 - 0.8z^{-1}} = \frac{z}{z - 0.8}$$

Solution Clearly, $H(z)$ has a zero at $z = 0$ and a pole at $p = 0.8$. Hence the frequency response of the system is

$$H(\omega) = \frac{e^{j\omega}}{e^{j\omega} - 0.8}$$

The magnitude response is

$$|H(\omega)| = \frac{|e^{j\omega}|}{|e^{j\omega} - 0.8|} = \frac{1}{\sqrt{1.64 - 1.6\cos\omega}}$$

and the phase response is

$$\theta(\omega) = \omega - \tan^{-1}\frac{\sin\omega}{\cos\omega - 0.8}$$

The magnitude and phase responses are illustrated in Fig. 4.43. Note that the peak of the magnitude response occurs at $\omega = 0$, the point on the unit circle closest to the pole located at 0.8.

If the magnitude response in (4.4.52) is expressed in decibels.

$$|H(\omega)|_{dB} = 20\log_{10}|b_0| + 20\sum_{k=1}^{M}\log_{10}V_k(\omega) - 20\sum_{k=1}^{N}\log_{10}U_k(\omega) \tag{4.4.58}$$

Thus the magnitude response is expressed as a sum of the magnitude factors in $|H(\omega)|$.

4.4.7 Input–Output Correlation Functions and Spectra

In Section 2.6.5 we developed several correlation relationships between the input and the output sequences of an LTI system. Specifically, we derived the following equations:

$$r_{yy}(m) = r_{hh}(m) * r_{xx}(m) \tag{4.4.59}$$

$$r_{yx}(m) = h(m) * r_{xx}(m) \tag{4.4.60}$$

where $r_{xx}(m)$ is the autocorrelation sequence of the input signal $\{x(n)\}$, $r_{yy}(m)$ is the autocorrelation sequence of the output $\{y(n)\}$, $r_{hh}(m)$ is the autocorrelation sequence of the impulse response $\{h(n)\}$, and $r_{yx}(m)$ is the crosscorrelation sequence between the output and the input signals. Since (4.4.59) and (4.4.60) involve the convolution operation, the z-transform of these equations yields

$$\begin{aligned} S_{yy}(z) &= S_{hh}(z)S_{xx}(z) \\ &= H(z)H(z^{-1})S_{xx}(z) \end{aligned} \tag{4.4.61}$$

$$S_{yx}(z) = H(z)S_{xx}(z) \tag{4.4.62}$$

If we substitute $z = e^{j\omega}$ in (4.4.62), we obtain

$$\begin{aligned} S_{yx}(\omega) &= H(\omega)S_{xx}(\omega) \\ &= H(\omega)|X(\omega)|^2 \end{aligned} \tag{4.4.63}$$

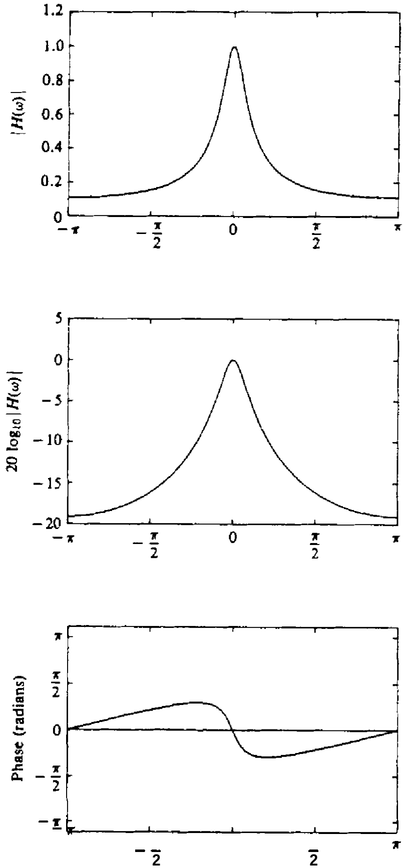


Figure 4.42 Magnitude and phase of system with $H(z) = 1/(1 - 0.8z^{-1})$.

where $S_{yx}(\omega)$ is the cross-energy density spectrum of $\{y(n)\}$ and $\{x(n)\}$. Similarly, evaluating $S_{yy}(z)$ on the unit circle yields the energy density spectrum of the output signal as

$$S_{yy}(\omega) = |H(\omega)|^2 S_{xx}(\omega) \tag{4.4.64}$$

where $S_{xx}(\omega)$ is the energy density spectrum of the input signal.

Since $r_{yy}(m)$ and $S_{yy}(\omega)$ are a Fourier transform pair. it follows that

$$r_{yy}(m) = \frac{1}{2\pi}\int_{-\pi}^{\pi} S_{yy}(\omega)e^{j\omega m}\,d\omega \tag{4.4.65}$$

The total energy in the output signal is simply

$$\begin{aligned} E_y &= \frac{1}{2\pi}\int_{-\pi}^{\pi} S_{yy}(\omega)d\omega = r_{yy}(0) \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} |H(\omega)|^2\, S_{xx}(\omega)d\omega \end{aligned} \tag{4.4.66}$$

The result in (4.4.66) may be used to easily prove that E, ≥ 0.

Finally, we note that if the input signal has a flat spectrum [i.e., $S_{xx}(\omega) = E_x$ = constant for $\pi \leq \omega \leq -\pi$], (4.4.63) reduces to

$$S_{yx}(\omega) = H(\omega)E_x \tag{4.4.67}$$

where E_x is the constant value of the spectrum. Hence

$$H(\omega) = \frac{1}{E_x} S_{yx}(\omega) \tag{4.4.68}$$

or, equivalently,

$$h(n) = \frac{1}{E_x} r_{yx}(m) \tag{4.4.69}$$

The relation in (4.4.69) implies that $h(n)$ can be determined by exciting the input to the system by a spectrally flat signal $\{x(n)\}$, and crosscorrelating the input with the output of the system. This method is useful in measuring the impulse response of an unknown system.

4.4.8 Correlation Functions and Power Spectra for Random Input Signals

This development parallels the derivations in Section **4.4.7.** with the exception that we now deal with statistical moments of the input and output signals of an LTI system. The various statistical parameters are introduced in Appendix A.

Let us consider a discrete-time linear time-invariant system with unit sample response $\{h(n)\}$ and frequency response $H(f)$. For this development we assume that $\{h(n)\}$ is real. Let $x(n)$ be a sample function of a stationary random process $X(n)$ that excites the system and let $y(n)$ denote the response of the system to $x(n)$.

From the convolution summation that relates the output to the input we have

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)x(n-k) \tag{4.4.70}$$

Since $x(n)$ is a random input signal, the output is also a random sequence. In other words, for each sample sequence $x(n)$ of the process $X(n)$, there is a corresponding sample sequence $y(n)$ of the output random process $Y(n)$. We wish to relate the statistical characteristics of the output random process $Y(n)$ to the statistical characterization of the input process and the characteristics of the system.

The expected value of the output $y(n)$ is

$$m_y \equiv E[y(n)] = E[\sum_{k=-\infty}^{\infty} h(k)x(n-k)]$$

$$= \sum_{k=-\infty}^{\infty} h(k)E[x(n-k)] \tag{4.4.71}$$

$$m_y = m_x \sum_{k=-\infty}^{\infty} h(k)$$

From the Fourier transform relationship

$$H(\omega) = \sum_{k=-\infty}^{\infty} h(k)e^{-j\omega k} \tag{4.4.72}$$

we have

$$H(0) = \sum_{k=-\infty}^{\infty} h(k) \tag{4.4.73}$$

which is the dc gain of the system. The relationship in **(4.4.73)** allows us to express the mean value in **(4.4.71)** as

$$m_y = m_x H(0) \tag{4.4.74}$$

The autocorrelation sequence for the output random process is

$$\begin{aligned}\gamma_{yy}(m) &= E[y^*(n)y(n+m)] \\ &= E\left[\sum_{k=-\infty}^{\infty} h(k)x^*(n-k) \sum_{j=-\infty}^{\infty} h(j)x(n+m-j)\right] \\ &= \sum_{k=-\infty}^{\infty}\sum_{j=-\infty}^{\infty} h(k)h(j)E[x^*(n-k)x(n+m-j)] \\ &= \sum_{k=-\infty}^{\infty}\sum_{j=-\infty}^{\infty} h(k)h(j)\gamma_{xx}(k-j+m)\end{aligned} \tag{4.4.75}$$

This is the general form for the autocorrelation of the output in terms of the autocorrelation of the input and the impulse response of the system.

A special form of **(4.4.75)** is obtained when the input random process is white, that is, when $m_x = 0$ and

$$\gamma_{xx}(m) = \sigma_x^2\delta(m) \tag{4.4.76}$$

where $\sigma_x^2 \equiv \gamma_{xx}(0)$ **is** the input signal power. Then **(4.4.75)** reduces to

$$\gamma_{yy}(m) = \sigma_x^2 \sum_{k=-\infty}^{\infty} h(k)h(k+m) \tag{4.4.77}$$

Under this condition the output process has the average power

$$\gamma_{yy}(0) = \sigma_x^2 \sum_{n=-\infty}^{\infty} h^2(n) = \sigma_x^2 \int_{-1/2}^{1/2} |H(f)|^2 df \tag{4.4.78}$$

where **we** have applied Parseval's theorem, The relationship in **(4.4.75)** can **be** transformed into the frequency domain by determining the power density spectrum of $\gamma_{yy}(m)$. We have

$$\begin{aligned}
\Gamma_{yy}(\omega) &= \sum_{m=-\infty}^{\infty} \gamma_{yy}(m)e^{-j\omega m} \\
&= \sum_{m=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} h(k)h(l)\gamma_{xx}(k-l+m) \right] e^{-j\omega m} \\
&= \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} h(k)h(l) \left[\sum_{m=-\infty}^{\infty} \gamma_{xx}(k-l+m)e^{-j\omega m} \right] \\
&= \Gamma_{xx}(f) \left[\sum_{k=-\infty}^{\infty} h(k)e^{j\omega k} \right] \left[\sum_{l=-\infty}^{\infty} h(l)e^{-j\omega l} \right] \\
&= |H(\omega)|^2 \Gamma_{xx}(\omega)
\end{aligned} \tag{4.4.79}$$

This is the desired relationship for the power density spectrum of the output process, in terms of the power density spectrum of the input process and the frequency response of the system.

The equivalent expression for continuous-time systems with random inputs is

$$\Gamma_{yy}(F) = |H(F)|^2 \Gamma_{xx}(F) \tag{4.4.80}$$

where the power density spectra $\Gamma_{yy}(F)$ and $\Gamma_{xx}(F)$ are the Fourier transforms of the autocorrelation functions $\gamma_{yy}(\tau)$ and $\gamma_{xx}(\tau)$, respectively, and where $H(F)$ is the frequency response of the system, which is related to the impulse response by the Fourier transform. that is,

$$H(F) = \int_{-\infty}^{\infty} h(t)e^{-j2\pi Ft}\,dt \tag{4.4.81}$$

As a final exercise, we determine the crosscorrelation of the output $y(n)$ with the input signal $x(n)$. If we multiply both sides of (4.4.70) by $x^*(n-m)$ and take the expected value, we obtain

$$E[y(n)x^*(n-m)] = E\left[\sum_{k=-\infty}^{\infty} h(k)x^*(n-m)x(n-k) \right]$$

$$\begin{aligned}
\gamma_{yx}(m) &= \sum_{k=-\infty}^{\infty} h(k)E[x^*(n-m)x(n-k)] \\
&= \sum_{k=-\infty}^{\infty} h(k)\gamma_{xx}(m-k)
\end{aligned} \tag{4.4.82}$$

Since **(4.4.82)** has the form of a convolution, **the** frequency-domain equivalent expression is

$$\Gamma_{yx}(\omega) = H(\omega)\Gamma_{xx}(\omega) \tag{4.4.83}$$

In the special case where $x(n)$ is white noise, (4.4.83) reduces to

$$\Gamma_{yx}(\omega) = \sigma_x^2 H(\omega) \tag{4.4.84}$$

where σ_x^2 is the input noise power. This result means that an unknown system with frequency response $H(\omega)$ can be identified by exciting the input with white noise, crosscorrelating the input sequence with the output sequence to obtain $\gamma_{yx}(m)$, and finally, computing the Fourier transform of $\gamma_{yx}(m)$. The result of these computations is proportional to $H(\omega)$.

4.5 LINEAR TIME-INVARIANT SYSTEMS AS FREQUENCY-SELECTIVE FILTERS

The term filter is commonly used to describe a device that discriminates, according to some attribute of the objects applied at its input, what passes through it, For example, an air filter allows air to pass through it but prevents dust particles that are present in the air from passing through. An oil filter performs a similar function, with the exception that oil is the substance allowed to pass through the filter, while particles of dirt are collected at the input to the filter and prevented from passing through. In photography. an ultraviolet filter is often used to prevent ultraviolet light, which is present in sunlight and which is not a part of visible light, from passing through and affecting the chemicals on the film.

As we have observed in the preceding section, a linear time-invariant system also performs a type of discrimination or filtering among the various frequency components at its input. The nature of this filtering action is determined by the frequency response characteristics $H(\omega)$, which in turn depends on the choice of the system parameters (e.g., the coefficients $\{a_k\}$ and $\{b_k\}$ in the difference equation characterization of the system). Thus, by proper selection of the coefficients, we can design frequency-selective filters that pass signals with frequency components in some bands while they attenuate signals containing frequency components in other frequency bands.

In general, a linear time-invariant system modifies the input signal spectrum $X(\omega)$ according to its frequency response $H(\omega)$ to yield an output signal with spectrum $Y(\omega) = H(\omega)X(\omega)$. In a sense, $H(\omega)$ acts as a ***weighting function*** or a ***spectral shaping function*** to the different frequency components in the input signal. When viewed in this context, any linear time-invariant system can be considered to be a frequency-shaping filter, even though it may not necessarily completely block any or all frequency components. Consequently, the terms "linear time-invariant system" and "filter" are synonymous and are often used interchangeably.

We use the term ***filter*** to describe a linear time-invariant system used to perform spectral shaping or frequency-selective filtering. Filtering is used in digital signal processing in a variety of ways. For example, removal of undesirable noise from desired signals, spectral shaping such as equalization of communication channels, signal detection in radar, sonar, and communications, and for performing spectral analysis of signals, and so on.

4.5.1 Ideal Filter Characteristics

Filters are usually classified according to their frequency-domain characteristics as lowpass, highpass. bandpass. and bandstop or band-elimination filters. The ideal magnitude response characteristics of these types of filters are illustrated in Fig. 4.43. As shown. these ideal filters have a constant-pain (usually taken as unity-gain) passband characteristic and zero gain in their stopband.

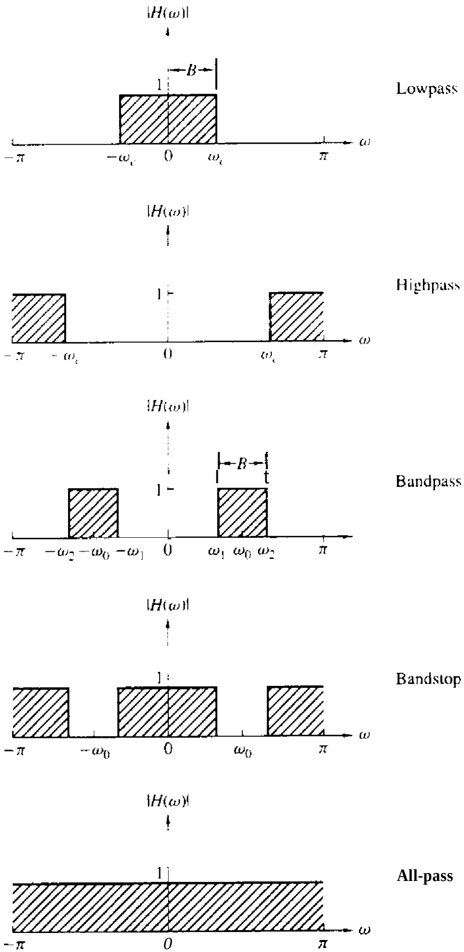


Figure 4.43 Magnitude responses for some ideal frequency-selective discrete-time filters.

Another characteristic of an ideal filter is a linear phase response. To demonstrate this point, let us assume that a signal sequence $\{x(n)\}$ with frequency components confined to the frequency range $\omega_1 < \omega < \omega_2$ is passed through a filter with frequency response

$$H(\omega) = \begin{cases} Ce^{-j\omega n_0}, & \omega_1 < \omega < \omega_2 \\ 0, & \text{otherwise} \end{cases} \tag{4.5.1}$$

where C and n_0 are constants. The signal at the output of the filter has a spectrum

$$\begin{aligned} Y(\omega) &= X(\omega)H(\omega) \\ &= CX(\omega)e^{-j\omega n_0} \qquad \omega_1 < \omega < \omega_2 \end{aligned} \tag{4.5.2}$$

By applying the scaling and time-shifting properties of the Fourier transform, we obtain the time-domain output

$$y(n) = Cx(n - n_0) \tag{4.5.3}$$

Consequently, the filter output is simply a delayed and amplitude-scaled version of the input signal. A pure delay is usually tolerable and is not considered a distortion of the signal. Neither is amplitude scaling. Therefore. ideal filters have a linear phase characteristic within their passhand. that is.

$$\Theta(\omega) = -\omega n_0 \tag{4.5.4}$$

The derivative of the phase with respect to frequency has the units of delay. Hence we can define the signal delay as a function of frequency as

$$\tau_g(\omega) = -\frac{d\Theta(\omega)}{d\omega} \tag{4.5.5}$$

$\tau_g(\omega)$ is usually called the ***envelope*** delay or the group *delay* of the filter. We interpret $\tau_g(\omega)$ as the time delay that a signal component of frequency ω undergoes as it passes from the input to the output of the system. Note that when $\Theta(\omega)$ is linear as in (4.5.4), $\tau_g(\omega) = n_0 =$ constant. In this case all frequency components of the input signal undergo the same time delay.

In conclusion, ideal filters have a constant magnitude characteristic and a linear phase characteristic within their passband. In all cases, such filters are not physically realizable but serve as a mathematical idealization of practical filters. For example, the ideal lowpass filter has an impulse response

$$h_{lp}(n) = \frac{\sin \omega_c \pi n}{\pi n} \qquad -\infty < n < \infty \tag{4.5.6}$$

We note that this filter is not causal and it is not absolutely summable and therefore it is also unstable. Consequently, this ideal filter is physically unrealizable. Nevertheless, its frequency response characteristics can be approximated very closely by practical, physically realizable filters, as will be demonstrated in Chapter 8.

In the following discussion, we treat the design of some simple digital filters by the placement of poles and zeros in the z-plane. We have already described how the location of poles and zeros affects the frequency response characteristics of the system. In particular. in Section 4.4.6 we presented a graphical method for computing the frequency response characteristics from the pole–zero plot. This same approach can be used to design a number of simple but important digital filters with desirable frequency response characteristics.

The basic principle underlying the pole–zero placement method is to locate poles near points of the unit circle corresponding to frequencies to be emphasized. and to place zeros near the frequencies to be deemphasized. Furthermore, the following constraints must be imposed:

1. All poles should be placed inside the unit circle in order for the filter to be stable. However. zeros can be placed anywhere in the z-plane.
2. All complex zeros and poles must occur in complex-conjugate pairs in order for the filter coefficients to be real.

From our previous discussion we recall that for a given pole–zero pattern. the system function $H(z)$ can be expressed as

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}} = b_0 \frac{\prod_{k=1}^{M}(1 - z_k z^{-1})}{\prod_{k=1}^{N}(1 - p_k z^{-1})} \tag{4.5.7}$$

where b_0 is a gain constant selected to normalize the frequency response at some specified frequency. That is, b_0 is selected such that

$$|H(\omega_0)| = 1 \tag{4.5.8}$$

where ω_0 is a frequency in the passband of the filter. Usually, N is selected to equal or exceed M. so that the filter has more nontrivial poles than zeros.

In the next section. we illustrate the method of pole–zero placement in the design of some simple lowpass. highpass. and bandpass filters. digital resonators. and comb filters. The design procedure is facilitated when carried out interactively on a digital computer with a graphics terminal.

4.5.2 Lowpass, Highpass, and Bandpass Filters

In the design of lowpass digital filters. the poles should be placed near the unit circle at points corresponding to low frequencies (near $\omega = 0$) and zeros should be placed near or on the unit circle at points corresponding to high frequencies (near $\omega = \pi$). The opposite holds true for highpass filters.

Figure 4.44 illustrates the pole–zero placement of three lowpass and three highpass filters. The magnitude and phase responses for the single-pole filter with system function

$$H_1(z) = \frac{1 - a}{1 - az^{-1}} \tag{4.5.9}$$

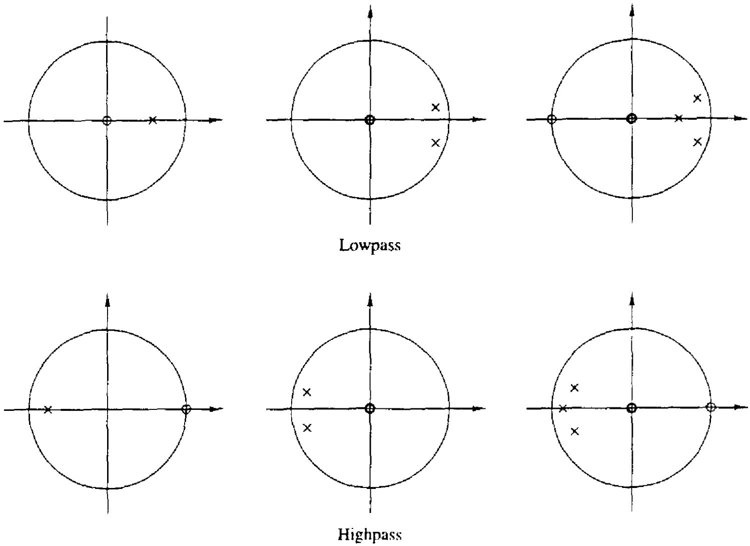


Figure 4.44 Pole–zero patterns for *several* lowpass and highpass filters.

are illustrated in Fig. 4.45 for $a = 0.9$. The gain G was selected as $1 - a$, so that the filter has unity gain at $\omega = 0$. The gain of this filter at high frequencies is relatively small.

The addition of a zero at $z = -1$ further attenuates the response of the filter at high frequencies. This leads to a filter with a system function

$$H_2(z) = \frac{1-a}{2} \frac{1+z^{-1}}{1-az^{-1}} \tag{4.5.10}$$

and a frequency response characterstic that is also illustrated in Fig. 4.45. In this case the magnitude of $H_2(\omega)$ goes to zero at $\omega = \pi$.

Similarly, we can obtain simple highpass filters by reflecting (folding) the pole–zero locations of the lowpass filters about the imaginary axis in the z-plane. Thus we obtain the system function

$$H_3(z) = \frac{1-a}{2} \frac{1-z^{-1}}{1+az^{-1}} \tag{4.5.11}$$

which has the frequency response characteristics illustrated in Fig. 4.46 for $a = 0.9$.

Example 45.1

A two-pole lowpass filter has the system function

$$H(z) = \frac{b_0}{(1-pz^{-1})^2}$$

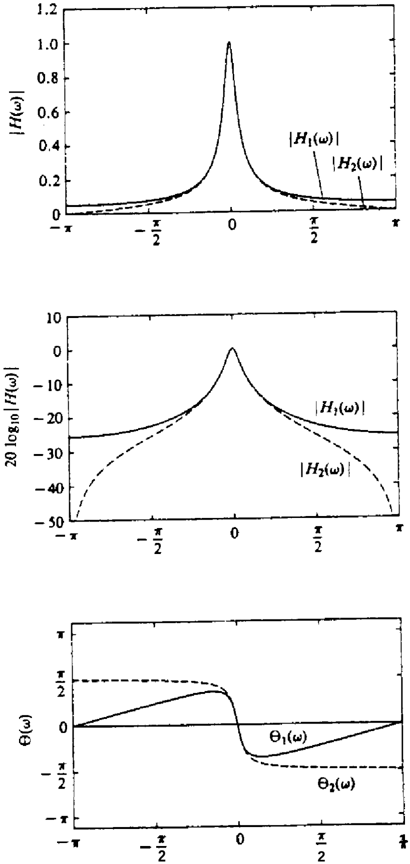


Figure 4.45 Magnitude and phase response of (1) a single-pole filter and (2) a one-pole. one-zero filter; $H_1(z) = (1-a)/(1-az^{-1})$, $H_2(z) = [(1-a)/2][(1+z^{-1})/(1-az^{-1})]$ **and a** $= 0.9$.

Determine the values of b_0 and p such that the frequency response $H(\omega)$ satisfies the conditions

$$H(0) = 1$$

and

$$\left|H\left(\frac{\pi}{4}\right)\right|^2 = \frac{1}{2}$$

Solution **At $\omega = 0$ we have**

$$H(0) = \frac{b_0}{(1-p)^2} - 1$$

Hence

$$b_0 = (1-p)^2$$

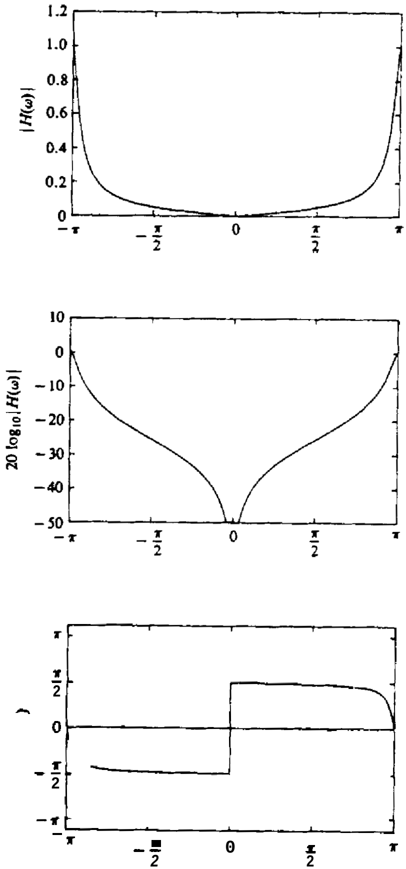


Figure 4.46 Magnitude and phase response of a simple highpass filter: $H(z) = [(1-a)/2][(1-z^{-1})/(1+az^{-1})]$ with $a = 0.9$.

At $\omega = \pi/4$,

$$H\left(\frac{\pi}{4}\right) = \frac{(1-p)^2}{(1-pe^{-j\pi/4})^2}$$

$$= \frac{(1-p)^2}{(1-p\cos(\pi/4) + jp\sin(\pi/4))^2}$$

$$= \frac{(1-p)^2}{(1-p/\sqrt{2}+jp/\sqrt{2})^2}$$

Hence

$$\frac{(1-p)^4}{[(1-p/\sqrt{2})^2+p^2/2]^2} = \frac{1}{2}$$

or, equivalently,

$$\sqrt{2}(1-p)^2 = 1 + p^2 - \sqrt{2}p$$

The value of $p = 0.32$ satisfies this equation. Consequently, the system function for the desired filter is

$$H(z) = \frac{0.46}{(1-0.32z^{-1})^2}$$

The same principles can be applied for the design of bandpass filters. Basically, the bandpass filter should contain one or more pairs of complex-conjugate poles near the unit circle, in the vicinity of the frequency band that constitutes the passband of the filter. The following example serves to illustrate the basic ideas.

Example 4.5.2

Design a two-pole bandpass filler that has the center of its passband ar $\omega = \pi/2$, zero in its frequency response characteristic at $\omega = 0$ and $\omega = \pi$, and its magnitude response is $1/\sqrt{2}$ at $\omega = 4\pi/9$.

Solution Clearly, the filter must have poles at

$$p_{1,2} = re^{\pm j\pi/2}$$

and zeros at $z = 1$ and $z = -1$. Consequently, the system function is

$$\begin{aligned} H(z) &= G\frac{(z-1)(z+1)}{(z-jr)(z+jr)} \\ &= G\frac{z^2-1}{z^2+r^2} \end{aligned}$$

The gain factor is determined by evaluating the frequency response $H(\omega)$ of the filter at $\omega = \pi/2$. Thus we have

$$\begin{aligned} H\left(\frac{\pi}{2}\right) &= G\frac{2}{1-r^2} = 1 \\ G &= \frac{1-r^2}{2} \end{aligned}$$

The value of r is determined by evaluating $H(\omega)$ at $\omega = 4\pi/9$. Thus we have

$$\left|H\left(\frac{4\pi}{9}\right)\right|^2 = \frac{(1-r^2)^2}{4}\frac{2-2\cos(8\pi/9)}{1+r^4+2r^2\cos(8\pi/9)} = \frac{1}{2}$$

or, equivalently,

$$1.94(1-r^2)^2 = 1 - 1.88r^2 + r^4$$

The value of $r^2 = \mathbf{0.7}$ satisfies this equation. Therefore, the system function for the desired filter is

$$H(z) = 0.15\frac{1-z^{-2}}{1+0.7z^{-2}}$$

Its frequency response is illustrated in Fig. **4.47.**

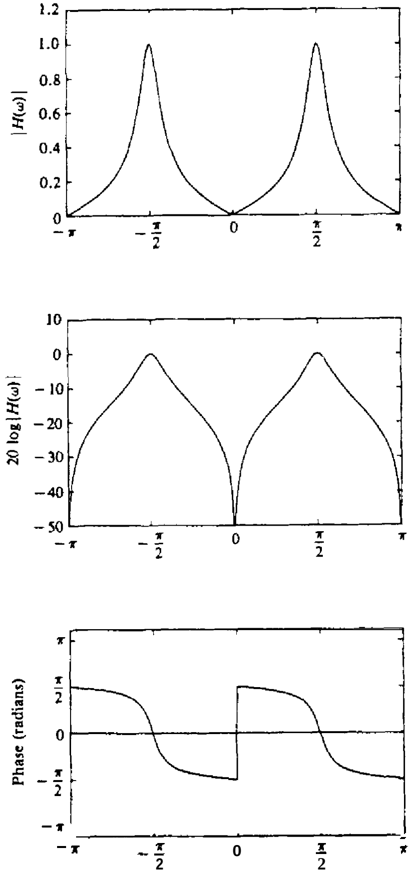


Figure 4.47 Magnitude and phase response of a simple bandpass filter in Example 4.5.2: $H(z) = 0.15[(1 - z^{-2})/(1 + 0.7z^{-2})]$.

It should be emphasized that the main purpose of the foregoing methodology for designing simple digital filters by pole–zero placement is to provide insight into the effect that poles and zeros have on the frequency response characteristic of systems. The methodology is not intended as a good method for designing digital fitters with well-specified passband and stopband characteristics. Systematic methods for the design of sophisticated digital filters for practical applications are discussed in Chapter 8.

A simple lowpass-to-highpass filter transformation. Suppose that we have designed a prototype lowpass filter with impulse response $h_{lp}(n)$. By using the frequency translation property of the Fourier transform. it is possible to convert the prototype filter to either a bandpass or a highpass filter. Frequency transformations for converting a prototype lowpass filter into a filter of another type are described in detail in Section 8.3. In this section we present a simple-frequency transformation for converting a lowpass filter into a highpass filter. and vice versa.

If $h_{lp}(n)$ denotes the impulse response of a lowpass filter with frequency response $H_{lp}(\omega)$. a highpass filter can be obtained by translating $H_{lp}(\omega)$ by π radians (i.e.. replacing ω by $\omega - \pi$). Thus

$$H_{hp}(\omega) = H_{lp}(\omega - \pi) \tag{4.5.12}$$

where $H_{hp}(\omega)$ is the frequency response of the highpass filter. Since a frequency translation of π radians is equivalent to multiplication of the impulse response $h_{lp}(n)$ by $e^{j\pi n}$. the impulse response of the highpass filter is

$$h_{hp}(n) = (e^{j\pi})^n h_{lp}(n) = (-1)^n h_{lp}(n) \tag{4.5.13}$$

Therefore. the impulse response of the highpass filter is simply obtained from the impulse response of the lowpass filter by changing the signs of the odd-numbered samples in $h_{lp}(n)$. Conversely.

$$h_{lp}(n) = (-1)^n h_{hp}(n) \tag{4.5.14}$$

If tho lowpass filter is described by the difference equation

$$y(n) = -\sum_{k=1}^{N} a_k y(n-k) + \sum_{k=0}^{M} b_k x(n-k) \tag{4.5.15}$$

its frequency response is

$$H_{lp}(\omega) = \frac{\displaystyle\sum_{k=0}^{M} b_k e^{-j\omega k}}{1 + \displaystyle\sum_{k=1}^{N} a_k e^{-j\omega k}} \tag{4.5.16}$$

Now. if we replace ω by $\omega - \pi$, in (4.5.16). then

$$H_{hp}(\omega) = \frac{\displaystyle\sum_{k=0}^{M} (-1)^k b_k e^{-j\omega k}}{1 + \displaystyle\sum_{k=1}^{N} (-1)^k a_k e^{-j\omega k}} \tag{4.5.17}$$

which corresponds to the difference equation

$$y(n) = -\sum_{k=1}^{N} (-1)^k a_k y(n-k) + \sum_{k=0}^{M} (-1)^k b_k x(n-k) \tag{4.5.18}$$

Example 45.3

Convert the lowpass filter described by the difference equation

$$y(n) = 0.9y(n-1) + 0.1x(n)$$

into a highpass filter.

Solution The difference equation for the highpass filter, according to (4.5.18), is

$$y(n) = -0.9y(n-1) + 0.1x(n)$$

and its frequency response is

$$H_{hp}(\omega) = \frac{0.1}{1 + 0.9e^{-j\omega}}$$

The reader may **verify** that $H_{hp}(\omega)$ is indeed highpass.

4.5.3 Digital Resonators

A *digital* resonator is a special two-pole bandpass filter with the pair of complex-conjugate poles located near the unit circle as shown in Fig. 4.48(a). The magnitude of the frequency response of the filter is shown in Fig. 4.48(b). The name resonator refers to the fact that the filter has a large magnitude response (i.e., it resonates) in the vicinity of the pole location. The angular position of the pole determines the resonant frequency of the filter. Digital resonators are useful in many applications, including simple bandpass filtering and speech generation.

In the design of a digital resonator with a resonant peak at or near $\omega = \omega_0$, we select the complex-conjugate poles at

$$p_{1,2} = re^{\pm j\omega_0} \qquad 0 < r < 1$$

In addition, we can select up to two zeros. Although there are many possible choices, two cases are of special interest. One choice is to locate the zeros at the origin. The other choice is to locate a zero at $z = 1$ and a zero at $z = -1$. This choice completely eliminates the response of the filter at frequencies $\omega = 0$ and $\omega = \pi$, and it is useful in many practical applications.

The system function of the digital resonator **with** zeros at the origin is

$$H(z) = \frac{b_0}{(1 - re^{j\omega_0}z^{-1})(1 - re^{-j\omega_0}z^{-1})} \tag{4.5.19}$$

$$H(z) = \frac{b_0}{1 - (2r\cos\omega_0)z^{-1} + r^2z^{-2}} \tag{4.5.20}$$

Since $|H(\omega)|$ has its peak at or near $\omega = \omega_0$, we select the gain b_0 so that $|H(\omega_0)| = 1$. From (4.5.19) we obtain

$$\begin{aligned} H(\omega_0) &= \frac{b_0}{(1 - re^{j\omega_0}e^{-j\omega_0})(1 - re^{-j\omega_0}e^{-j\omega_0})} \\ &= \frac{b_0}{(1-r)(1 - re^{-j2\omega_0})} \end{aligned} \tag{4.5.21}$$

and hence

$$|H(\omega_0)| = \frac{b_0}{(1-r)\sqrt{1 + r^2 - 2r\cos 2\omega_0}} = 1$$

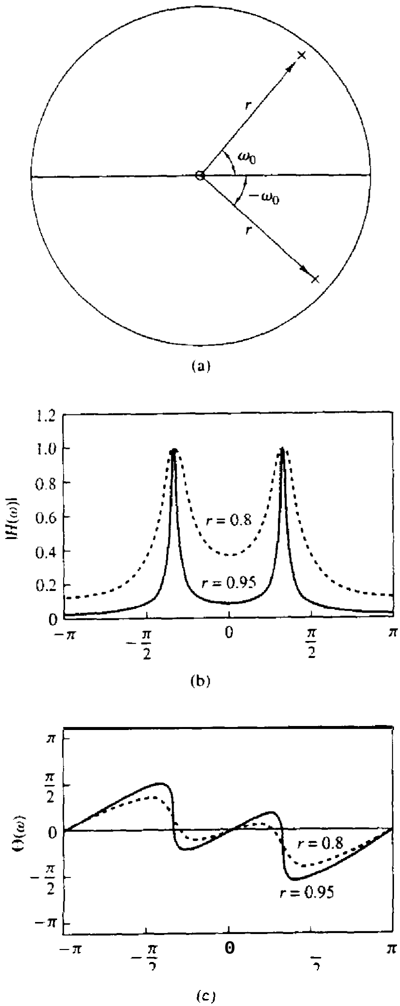


Figure 4.48 (a) Pole-zero pattern and (b) the corresponding magnitude and phase response of a digital resonator with (1) $r = 0.8$ and (2) $r = 0.95$.

Thus the desired normalization factor is

$$b_0 = (1-r)\sqrt{1+r^2-2r\cos 2\omega_0} \tag{4.5.22}$$

The frequency response of the resonator in (4.5.19) can be expressed as

$$\begin{aligned} |H(\omega)| &= \frac{b_0}{U_1(\omega)U_2(\omega)} \\ \Theta(\omega) &= 2\omega - \Phi_1(\omega) - \Phi_2(\omega) \end{aligned} \tag{4.5.23}$$

where $U_1(\omega)$ and $U_2(\omega)$ are the magnitudes of the vectors from p_1 and p_2 to the point ω in the unit circle and $\Phi_1(\omega)$ and $\Phi_2(\omega)$ are the corresponding angles of these two vectors. The magnitudes $U_1(\omega)$ and $U_2(\omega)$ may be expressed as

$$\begin{aligned} U_1(\omega) &= \sqrt{1+r^2-2r\cos(\omega_0-\omega)} \\ U_2(\omega) &= \sqrt{1+r^2-2r\cos(\omega_0+\omega)} \end{aligned} \tag{4.5.24}$$

For any value of r, $U_1(\omega)$ takes its minimum value $(1-r)$ at $\omega = \omega_0$. The product $U_1(\omega)U_2(\omega)$ reaches a minimum value at the frequency

$$\omega_r = \cos^{-1}\left(\frac{1+r^2}{2r}\cos\omega_0\right) \tag{4.5.25}$$

which defines precisely the resonant frequency of the filter. We observe that when r is very close to unity, $\omega_r \approx \omega_0$, which is the angular position of the pole. We also observe that as r approaches unity, the resonance peak becomes sharper because $U_1(\omega)$ changes more rapidly in relative size in the vicinity of ω_0. A quantitative measure of the sharpness of the resonance is provided by the 2-dB bandwidth $\Delta\omega$ of the filter. For values of r close to unity,

$$\Delta\omega \approx 2(1-r) \tag{4.5.26}$$

Figure 4.48 illustrates the magnitude and phase of digital resonators with $\omega_0 = \pi/3$, $r = 0.8$ and $\omega_0 = \pi/3$, $r = 0.95$. We note that the phase response undergoes its greatest rate of change near the resonant frequency.

If the zeros of the digital resonator are placed at $z = 1$ and $z = -1$, the resonator has the system function

$$\begin{aligned} H(z) &= G\frac{(1-z^{-1})(1+z^{-1})}{(1-re^{j\omega_0}z^{-1})(1-re^{-j\omega_0}z^{-1})} \\ &= G\frac{1-z^{-2}}{1-(2r\cos\omega_0)z^{-1}+r^2z^{-2}} \end{aligned} \tag{4.5.27}$$

and a frequency response characteristic

$$H(\omega) = b_0\frac{1-e^{-j2\omega}}{[1-re^{j(\omega_0-\omega)}][1-re^{-j(\omega_0+\omega)}]} \tag{4.5.28}$$

We observe that the zeros at $z = \pm 1$ affect both the magnitude and phase response of the resonator. For example, the magnitude response is

$$|H(\omega)| = b_0\frac{N(\omega)}{U_1(\omega)U_2(\omega)} \tag{4.5.29}$$

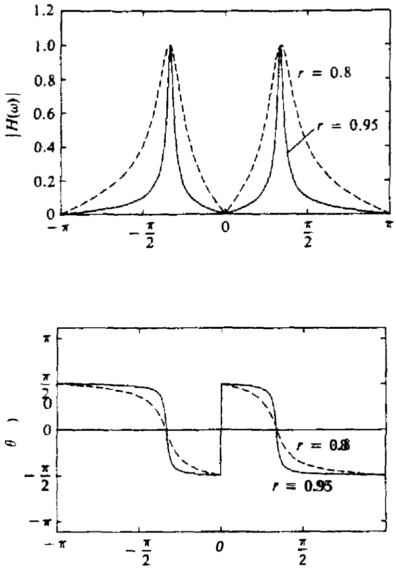


Figure 4.49 Magnitude and phase response of digital resonator with zeros at $\omega = 0$ and $\omega = \pi$ and (1) $r = 0.8$ and (2) $r = 0.95$.

where $N(\omega)$ is defined as

$$N(\omega) = \sqrt{2(1 - \cos 2\omega)}$$

Due to the presence of the zero factor. the resonant frequency is altered from that given by the expression in (4.5.25). The bandwidth of the filter is also altered. Although exact values for these two parameters are rather tedious to derive, we can easily compute the frequency response in (4.5.28) and compare the result with the previous case in which the zeros are located at the origin.

Figure 4.49 illustrates the magnitude and phase characteristics for $\omega_0 = \pi/3$, r $= 0.8$ and $\omega_0 = \pi/3$, $r = 0.95$. We observe that this filter has a slightly smaller bandwidth than the resonator, which has zeros at the origin. In addition. there appears to be a very small shift in the resonant frequency due to the presence of the zeros.

4.5.4 Notch Filters

A notch filter is a filter that contains one or more deep notches or, ideally, perfect nulls in its frequency response characteristic. Figure 4.50 illustrates the frequency response characteristic of a notch filter with nulls at frequencies ω_0 and ω_1. Notch filters are useful in many applications where specific frequency components must be eliminated. For example, instrumentation and recording systems require that the power-line frequency of 60 Hz and its harmonics be eliminated.